HSC Mathematics Extension 1: Combinatorics Practice

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Download PDF and resources:
`https://github.com/vuhung16au/math-olympiad-ml`

Mathematics may be defined as the economy of counting. There is no problem in the whole of mathematics which cannot be solved by direct counting.

Ernst Mach

Contents

1 Introduction

1.1 Project Overview

This booklet is a growing collection of combinatorics problems for NSW HSC Mathematics Extension 1 students. It now covers a wider range of syllabus ideas including permutations, combinations, Pascal's triangle, circular arrangements, committee selection, simple probability, expected value, the pigeonhole principle, geometric counting, and proof-style binomial identities.

1.2 Target Audience

This resource is written for students who want extra practice with combinatorics in a clear, classroom-friendly format. The solutions are intentionally brief, but each one points out the key counting idea behind the answer.

1.3 How to Use This Booklet

Read the short summary of the core formulas first, then attempt each problem before reading the solution. As you work, decide whether each question is about selection or arrangement, look for structure such as gaps in circles, repeated objects, complementary counting, or balanced groups, and use the takeaways to build pattern recognition for future HSC questions.

1.4 Extension 1 Knowledge Needed

Most problems in this booklet can be solved using standard HSC Mathematics Extension 1 ideas rather than advanced university mathematics. The key background knowledge is:

Title	**Explanation**
Factorials and notation	Understanding $n!$, permutation notation ${}^nP_r$, and combination notation $\binom{n}{r}$. For positive integer n, $n! = n(n-1)(n-2)\cdots 2\cdot 1$. Also, $0! = 1$, $1! = 1$, $2! = 2$, $3! = 6$, $4! = 24$, $5! = 120$. Here $\binom{n}{r}$ means the number of ways to choose r objects from n objects (order does not matter).
The multiplication principle	Breaking a problem into stages and multiplying the number of choices at each stage.
Order versus selection	Recognising when order matters (permutations) and when it does not (combinations).
Repeated objects	Dividing by repeated factorials when arranging words such as `COOKED` or `CONDOBOLIN`.
Circular permutations	Using $(n-1)!$ for arrangements around a circle and handling restrictions by fixing seats, using blocks, or using gaps.
Complementary counting	Counting all cases first, then subtracting the unwanted ones.
Casework and restrictions	Splitting problems into clean cases such as fixed numbers of girls and boys, men and women, or chosen leaders.
Pascal's triangle and binomial coefficients	Using identities such as $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$ and the symmetry rule $\binom{n}{r} = \binom{n}{n-r}$.

1.5 Core Formulas and Ideas

Fundamental Counting Principle. If a task is completed in stages with a choices for the first stage, b for the second, and c for the third, then the total number of outcomes is

$$a \times b \times c.$$

Generalized counting principle. Suppose that k experiments are to be performed, with the number of possible outcomes being n_i for the ith experiment. Then there are

$$n_1 \cdot n_2 \cdot \cdots \cdot n_k$$

possible outcomes of all k experiments together.

What is a set? A **set** is a collection of distinct objects. In a set, order does not matter, and repeated copies are not listed separately. For example,

$$\{1, 2, 3\} = \{3, 2, 1\},$$

and writing $\{1, 1, 2\}$ does not create a new 3-element set; as a set, it is just $\{1, 2\}$.

In counting problems, this usually means:

- the objects in the set are treated as distinct unless the question says some are equal or repeated
- repetition is *not* automatically allowed just because we are arranging or choosing from a set
- if repetition *is* allowed, the question should say so explicitly.

Permutations versus combinations. A **permutation** is an arrangement, so order matters. A **combination** is a selection, so order does not matter.

Worked example. Suppose 3 students are Alice, Ben, and Chloe.

- If we choose a captain and vice-captain, order matters, so this is a permutation:

  $$^3P_2 = 3 \times 2 = 6.$$

  For example, "Alice then Ben" is different from "Ben then Alice".

- If we simply choose a team of 2, order does not matter, so this is a combination:

  $$\binom{3}{2} = 3.$$

  The team {Alice, Ben} is the same team as {Ben, Alice}.

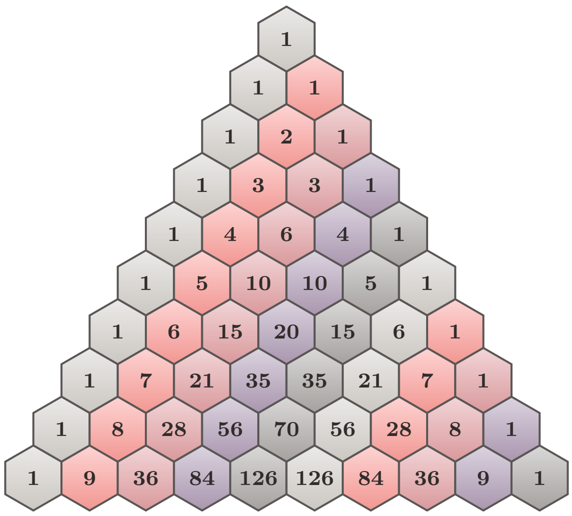
The following formulas are useful to keep together as a quick reference:

Name of formula	**Formula**
Permutations without repetition	${}^nP_r = \dfrac{n!}{(n-r)!}$
Combinations	$\binom{n}{r} = \dfrac{n!}{r!(n-r)!}$
Multinomial coefficients	If $n = a+b+c+\cdots$, then the number of distinct arrangements is $\dfrac{n!}{a!\,b!\,c!\cdots}$.
Multinomial expansion note	In $(x_1+x_2+\cdots+x_k)^m$, the coefficient of $x_1^{a_1}x_2^{a_2}\cdots x_k^{a_k}$ where $a_1+a_2+\cdots+a_k=m$ is $\dfrac{m!}{a_1!a_2!\cdots a_k!}$.
Circular permutations	$(n-1)!$
Pascal's triangle identity	$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$
Binomial symmetry	$\binom{n}{r} = \binom{n}{n-r}$

1.6 Pascal's Triangle

Pascal's triangle is a triangular arrangement of binomial coefficients. For HSC Extension 1, it is useful both for quick coefficient reading and for proving combinatorial identities.

For $n = 0, 1, 2, \ldots, 9$ (10 rows), Pascal's triangle can be shown with stacked hexagons:



Three facts to remember:

- **Pascal rule:**
$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}.$$
Each interior entry is the sum of the two entries above it.

- **Symmetry:**
$$\binom{n}{r} = \binom{n}{n-r}.$$
Each row reads the same from left to right and from right to left.

- **Row sum:**
$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n.$$
The sum of the entries in row n is a power of 2.

1.7 The Binomial Theorem

For a positive integer n,

$$(x+y)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 + \cdots + \binom{n}{n}y^n.$$

This theorem tells us how to expand powers such as $(x+y)^n$. The coefficients come directly from row n of Pascal's triangle.

Worked example. Expand $(x+2)^4$.

Using row 4 of Pascal's triangle, the coefficients are $1, 4, 6, 4, 1$:

$$\begin{aligned}(x+2)^4 &= \binom{4}{0}x^4 + \binom{4}{1}x^3(2) + \binom{4}{2}x^2(2^2) + \binom{4}{3}x(2^3) + \binom{4}{4}(2^4)\\ &= x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 16\\ &= x^4 + 8x^3 + 24x^2 + 32x + 16.\end{aligned}$$

This is especially useful later in the booklet when proving identities by comparing coefficients of matching powers of x.

Expected value. If a random variable takes values $x_1, x_2, \ldots$ with probabilities $p_1, p_2, \ldots,$ then

$$E(X) = x_1p_1 + x_2p_2 + \cdots .$$

1.8 Binomial Probability

Binomial probability is used when the same trial is repeated a fixed number of times and each trial has only two possible outcomes, often called *success* and *failure*.

The main features are:

- there are only two possible outcomes on each trial
- the probability of success is the same on every trial
- the number of trials is fixed
- the trials are independent

If X is the number of successes in n trials, and the probability of success on each trial is p, then

$$P(X=r) = \binom{n}{r}p^r(1-p)^{n-r}.$$

Here $\binom{n}{r}$ counts the number of ways to arrange the r successes among the n trials.

Example. A green counter is selected with probability 0.2 on each draw, and the draw is repeated 5 times with replacement. The probability of getting exactly 2 green counters is

$$P(X=2) = \binom{5}{2}(0.2)^2(0.8)^3.$$

This idea appears in repeated-draw questions later in the booklet.

1.9 The Pigeonhole Principle and Remainders

Suppose that n objects are placed into d boxes. If we divide n by d, we can write

$$n = qd + r, \qquad \text{where } 0 \le r < d.$$

This helps us read the pigeonhole principle more precisely:

- if $r = 0$, then at least one box contains at least q objects
- if $r > 0$, then at least one box contains at least $q + 1$ objects.

In particular, the smallest number of objects that guarantees at least one box contains at least $q + 1$ objects is

$$qd + 1.$$

This point of view is especially useful in "must happen" questions, where the boxes may be teams, remainders after division, topic choices, or vote totals.

1.10 Problem-Solving Habits for This Booklet

- Ask first: *Does order matter?*
- Check whether repetition is allowed or forbidden.
- In circle problems, decide whether to use a fixed reference point, a block, or a gap method.
- In identity questions, look for Pascal's triangle, symmetry, coefficient matching, or telescoping.
- In probability questions, decide whether the experiment is with replacement or without replacement.
- In "must happen" questions, test whether the pigeonhole principle or a contradiction argument applies.

Remark 1.1

The most common combinatorics mistake is using a permutation when the problem is actually a combination, or vice versa. A good first question is always: *Does order matter?*

Remark 1.2 (How fast counting grows)

A 19×19 Go board has 361 points. If each point is allowed to be black, white, or empty, then a rough count of possible board states is

$$3^{361} \approx 10^{172}.$$

Even after restricting to legal positions, the number is still about 10^{170}, which is unimaginably large. This is one of the best reminders that counting problems can grow far beyond ordinary intuition.

2 Part 1: Worked Problems

2.1 Basic

Problem 2.1: Four-digit numbers from 1 to 8

How many 4-digit numbers can you make using the digits 1 to 8 if no digit is repeated?

Solution 2.1

There are 8 choices for the thousands digit, then 7 choices for the hundreds digit, 6 for the tens digit, and 5 for the units digit.

$$8 \times 7 \times 6 \times 5 = 1680$$

So the number of such 4-digit numbers is $\boxed{1680}$.

Takeaways 2.1

- When order matters and repetition is not allowed, use a permutation count.
- The multiplication principle is often the fastest method for straightforward arrangement questions.

Problem 2.2: Rearranging the letters of COOKED

How many different arrangements of the letters in the word `COOKED` are possible?

Solution 2.2

The word `COOKED` has 6 letters in total, with:

$$O \text{ repeated 2 times.}$$

If all letters were distinct, there would be 6! arrangements. We divide by 2! for the repeated O's:

$$\frac{6!}{2!} = \frac{720}{2} = 360.$$

Hence the number of different arrangements is $\boxed{360}$.

Takeaways 2.2

- For repeated letters, start with the total factorial and divide by the factorial of each repeated group.
- This is the standard multiset permutation formula.

Problem 2.3: Choosing three coloured balls

A bag contains six different coloured balls.
How many different ways can three balls be selected?

Solution 2.3

Since the order of selection does not matter, we use combinations:

$$\binom{6}{3} = \frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} = 20.$$

So the number of different selections is $\boxed{20}$.

Takeaways 2.3

- Use combinations when order does not matter.
- The phrase "select two balls" usually signals a combination, not a permutation.

Problem 2.4: Rearranging the letters of CONDOBOLIN

In how many different ways can all the letters of the word `CONDOBOLIN` be arranged in a line?

Solution 2.4

The word `CONDOBOLIN` has 10 letters in total, with:

$$O \text{ repeated 3 times,} \qquad N \text{ repeated 2 times.}$$

If all letters were distinct, there would be 10! arrangements. We divide by 3! for the repeated O's and by 2! for the repeated N's:

$$\frac{10!}{3!\,2!} = \frac{3628800}{12} = 302400.$$

Hence the number of different arrangements is $\boxed{302400}$.

Takeaways 2.4

- This is another multiset permutation problem: start with $n!$ and divide by repeated factorials.
- Always count repeated letters carefully before substituting into the formula.

2.2 Medium

Problem 2.5: Seating around a circle

How many different ways can 9 people be seated in a circle if 2 people have already chosen their seats?

Solution 2.5

Once the two chosen seats are fixed, those two people no longer move. We only need to arrange the remaining 7 people in the remaining 7 seats.

$$7! = 5040$$

Therefore, the number of possible seatings is $\boxed{5040}$.

Takeaways 2.5

- Circular arrangements usually reduce by one factor because rotations are equivalent.
- If some seats are already fixed, the circular symmetry has already been broken.

Problem 2.6: Selections from URGOATED

Five letters are chosen from the letters of the word `URGOATED`, without ordering them. How many selections are there if the choice:

a) has no restriction?

b) must contain all four vowels?

c) must contain the letter `R`?

d) must contain the letter `U`, but not `D`?

Solution 2.6

All 8 letters in `URGOATED` are distinct: U, R, G, O, A, T, E, D.
a) No restriction:

$$\binom{8}{5} = 56$$

b) Must contain all four vowels U, O, A, E: We must choose 1 more letter from the remaining 4 consonants R, G, T, D.

$$\binom{4}{1} = 4$$

c) Must contain `R`: Fix `R`, then choose 4 more letters from the remaining 7 letters.

$$\binom{7}{4} = 35$$

d) Must contain `U`, but not `D`: Fix `U`, exclude `D`, then choose 4 more letters from R, G, O, A, T, E.

$$\binom{6}{4} = 15$$

So the answers are:

$$\boxed{56,\ 4,\ 35,\ 15}$$

Takeaways 2.6

- Restriction questions are often easiest if you first force required letters in, or remove forbidden letters.
- After handling the restriction, finish with a simple combination count.

Problem 2.7: Forming a Dota 2 e-sport team

a) How many different ways can a Dota 2 e-sport team of 5 students be chosen from a class of 30 students?

b) The team must have 2 girls and 3 boys. If the class contains 12 girls and 18 boys, how many teams are possible?

c) Suppose Vu and Zachary are both students in the class. Find the probability that both Vu and Zachary are chosen on the team.

d) Find the probability that neither Vu nor Zachary is chosen.

Solution 2.7

a) Choose any 5 students from 30:

$$\binom{30}{5} = 142506$$

b) Choose 2 girls from 12 and 3 boys from 18:

$$\binom{12}{2}\binom{18}{3} = 66 \times 816 = 53856$$

c) If Vu and Zachary must both be chosen, then choose the remaining 3 team members from the other 28 students:

$$\frac{\binom{28}{3}}{\binom{30}{5}} = \frac{3276}{142506} = \frac{2}{87}$$

So the probability is $\boxed{\frac{2}{87}}$.

d) If neither Vu nor Zachary is chosen, then all 5 team members must come from the other 28 students:

$$\frac{\binom{28}{5}}{\binom{30}{5}} = \frac{98280}{142506} = \frac{460}{667}$$

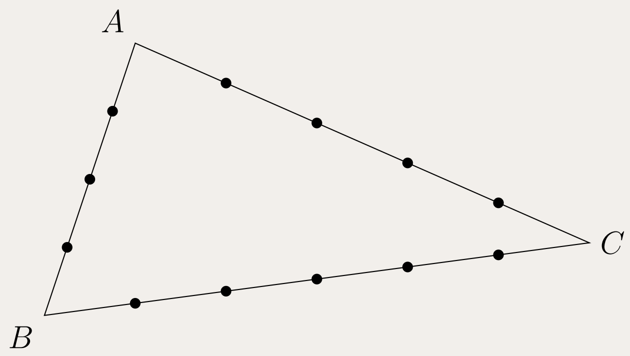
So the probability is $\boxed{\frac{460}{667}}$.

Takeaways 2.7

- For probabilities in equally likely team selections, count favourable teams over total teams.
- "Both chosen" means fix those people first; "neither chosen" means remove them from the pool.

Problem 2.8: Triangles from points on the sides of a triangle

The diagram shows triangle ABC with points chosen on each of the sides. On side AB, 3 points are chosen. On side AC, 4 points are chosen. On side BC, 5 points are chosen.



How many triangles can be formed using the chosen points as vertices?

Solution 2.8

There are

$$3 + 4 + 5 = 12$$

chosen points in total.
If no three points were collinear, the number of triangles would be

$$\binom{12}{3} = 220.$$

However, any 3 points chosen on the same side do not form a triangle.
On side AB, there are

$$\binom{3}{3} = 1$$

collinear triple.
On side AC, there are

$$\binom{4}{3} = 4$$

collinear triples.
On side BC, there are

$$\binom{5}{3} = 10$$

collinear triples.
So the number of valid triangles is

$$\binom{12}{3} - \binom{3}{3} - \binom{4}{3} - \binom{5}{3} = 220 - 1 - 4 - 10 = 205.$$

Hence the number of triangles is $\boxed{205}$.

Takeaways 2.8

- Start by counting all possible triples of points.
- Then subtract the triples that are collinear, since they do not form triangles.
- Geometric counting problems often reduce to combinations plus careful exclusion.

Problem 2.9: Age-limit check across teams

A sports association manages 13 junior teams. It decides to check the age of all players. Any team that has more than 3 players above the age limit will be penalised.
A total of 41 players are found to be above the age limit.
Will any team be penalised? Justify your answer.

Solution 2.9

Yes, at least one team must be penalised.
Suppose no team were penalised. Then each of the 13 teams would have at most 3 players above the age limit.
So the greatest possible total number of above-limit players would be

$$13 \times 3 = 39.$$

But the actual total is 41, and

$$41 > 39.$$

This is impossible if every team has at most 3 above-limit players.
Therefore, at least one team must have more than 3 players above the age limit, so at least one team will be penalised.

Takeaways 2.9

- This is a pigeonhole-principle style argument: if every group had the maximum allowed amount, the total would still be too small.
- A good way to prove that something *must* happen is to assume it does not, then show the numbers do not work.
- In combinatorics, "justify your answer" often means giving a short contradiction argument like this one.

Problem 2.10: Expected score with replacement

A game consists of randomly selecting 4 balls from a bag. After each ball is selected it is replaced in the bag. The bag contains 3 red balls and 7 green balls.
For each red ball selected, 10 points are earned and for each green ball selected, 5 points are deducted. For instance, if a player picks 3 red balls and 1 green ball, the score will be

$$3 \times 10 - 1 \times 5 = 25$$

points.
What is the expected score in the game?

Solution 2.10

For one draw,

$$P(\text{red}) = \frac{3}{10}, \qquad P(\text{green}) = \frac{7}{10}.$$

So the expected score from one draw is

$$\begin{aligned} E(\text{score per draw}) &= 10\left(\frac{3}{10}\right) + (-5)\left(\frac{7}{10}\right) \\ &= 3 - \frac{35}{10} \\ &= 3 - 3.5 \\ &= -0.5. \end{aligned}$$

Since there are 4 draws with replacement, the expected total score is

$$4 \times (-0.5) = -2.$$

Therefore, the expected score in the game is $\boxed{-2}$ points.

Takeaways 2.10

- With replacement, each draw has the same probabilities.
- The expected value of a total is the sum of the expected values of the individual draws.
- A negative expected value means that, on average, the game favours the house rather than the player.

Problem 2.11: Minimum votes needed to win

The members of a club voted for a new president. There were 15 candidates for the position of president and 3543 members voted. Each member voted for one candidate only.
One candidate received more votes than anyone else and so became the new president. What is the smallest number of votes the new president could have received?

Solution 2.11

To make the winner's vote total as small as possible, we want the other 14 candidates to receive as many votes as possible, while still having fewer votes than the winner.
Suppose the winner received x votes. Then each of the other 14 candidates can receive at most $x - 1$ votes.
So the greatest possible total number of votes is

$$x + 14(x - 1).$$

This must be at least 3543:

$$\begin{aligned} x + 14(x-1) &\geq 3543 \\ 15x - 14 &\geq 3543 \\ 15x &\geq 3557 \\ x &\geq \frac{3557}{15} = 237.13\ldots \end{aligned}$$

So the smallest possible integer value of x is

$$\boxed{238}.$$

This value is achievable, since if the winner gets 238 votes then the other 14 candidates can together receive up to

$$14 \times 237 = 3318$$

votes, and

$$3318 + 238 = 3556 > 3543,$$

so the 3543 votes can be distributed with the winner still strictly ahead.

Takeaways 2.11

- To minimize a maximum, try making all competitors as large as possible without overtaking it.
- This is a classic "worst-case" or extremal counting argument.
- After finding the bound, always check that it is actually achievable.

Problem 2.12: Selecting finalists and awarding places

Out of 10 contestants, six are to be selected for the final round of a competition. Four of those six will be placed 1st, 2nd, 3rd and 4th.
In how many ways can this process be carried out?

Solution 2.12

First choose the 6 finalists from the 10 contestants:

$$\binom{10}{6} = 210.$$

Now choose and order the 4 placed contestants from those 6 finalists:

$${}^6P_4 = 6 \times 5 \times 4 \times 3 = 360.$$

So the total number of ways is

$$\binom{10}{6} \times {}^6P_4 = 210 \times 360 = 75600.$$

Hence the process can be carried out in $\boxed{75600}$ ways.

Takeaways 2.12

- Break the process into stages: first select, then arrange.
- Use combinations for choosing the finalists because order does not matter there.
- Use permutations for the placings because 1st, 2nd, 3rd, and 4th are ordered positions.

Problem 2.13: Using the pigeonhole principle with topic choices

To complete a course, a student must choose and pass exactly three topics. There are eight topics from which to choose.
Last year 400 students completed the course.
Explain, using the pigeonhole principle, why at least eight students passed exactly the same three topics.

Solution 2.13

The number of different ways to choose exactly 3 topics from 8 is

$$\binom{8}{3} = 56.$$

So there are only 56 possible topic combinations.
Now distribute the 400 students among these 56 combinations. If at most 7 students had any one particular combination, then the total number of students could be at most

$$56 \times 7 = 392.$$

But in fact there are 400 students, and

$$400 > 392.$$

Therefore, by the pigeonhole principle, at least one of the 56 topic combinations must have been chosen by at least 8 students.
So at least eight students passed exactly the same three topics.

Takeaways 2.13

- Start by counting the number of possible categories, or “pigeonholes”.
- If the total number of objects exceeds the maximum possible without overlap, then one category must contain more than that maximum.
- In these questions, the pigeonhole principle often appears as: “If every group had at most k, the total would be too small.”

2.3 Advanced

Problem 2.14: Committee selection

A committee containing 4 men and 4 women is to be formed from a group of 10 men and 8 women.
In how many different ways can the committee be formed?

Solution 2.14

Choose the men and women independently:

$$\binom{10}{4}\binom{8}{4} = 210 \times 70 = 14700$$

Hence the number of possible committees is $\boxed{14700}$.

Takeaways 2.14

- When categories are separate, count each category and multiply.
- Committee questions are combination questions because order is irrelevant.

Problem 2.15: Pascal's triangle and a telescoping sum

i) The Pascal's triangle relation can be expressed as

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}.$$

Do not prove this relation. Show that

$$\binom{m}{R} = \binom{m+1}{R+1} - \binom{m}{R+1}.$$

ii) Hence, or otherwise, prove that

$$\binom{2000}{2000} + \binom{2001}{2000} + \binom{2002}{2000} + \cdots + \binom{2050}{2000} = \binom{2051}{2001}.$$

Solution 2.15

i) Apply Pascal's triangle with $n = m + 1$ and $r = R + 1$:

$$\binom{m+1}{R+1} = \binom{m}{R} + \binom{m}{R+1}.$$

Rearranging gives

$$\binom{m}{R} = \binom{m+1}{R+1} - \binom{m}{R+1}.$$

ii) Using part (i) with $R = 2000$,

$$\binom{m}{2000} = \binom{m+1}{2001} - \binom{m}{2001}.$$

Now let $m = 2000, 2001, \ldots, 2050$ and add:

$$\begin{aligned}\binom{2000}{2000} + \binom{2001}{2000} + \cdots + \binom{2050}{2000} &= \left(\binom{2001}{2001} - \binom{2000}{2001}\right) \\ &\quad + \left(\binom{2002}{2001} - \binom{2001}{2001}\right) \\ &\quad + \cdots \\ &\quad + \left(\binom{2051}{2001} - \binom{2050}{2001}\right).\end{aligned}$$

This is a telescoping sum, so all middle terms cancel. Also,

$$\binom{2000}{2001} = 0$$

because we cannot choose 2001 objects from 2000.
Hence the sum becomes

$$\binom{2051}{2001}.$$

Therefore,

$$\binom{2000}{2000} + \binom{2001}{2000} + \binom{2002}{2000} + \cdots + \binom{2050}{2000} = \boxed{\binom{2051}{2001}}.$$

Takeaways 2.15

- Pascal's triangle identities are often most useful after rearranging them into a subtraction form.
- A telescoping sum is one in which most terms cancel after expansion.
- In binomial coefficients, $\binom{n}{r} = 0$ when $r > n$, and that can clean up endpoint terms neatly.
- In an HSC proof question, full marks usually come from showing the key substitutions, the cancellation, and the final conclusion clearly, not just writing the final answer.
- Writing the process matters: markers need to see how part (i) is applied in part (ii), so short linking phrases such as "using part (i)" and "this is a telescoping sum" are worth including.

Remark 2.1 (Marking guide note)

For a question like this, students should read the marking guide as a list of *visible ingredients* their solution must contain. In part (i), a correct rearrangement with the substitution $n = m + 1$ and $r = R + 1$ is enough. In part (ii), full marks usually require more than the final identity: show how part (i) is applied to the general term, write out enough of the expanded sum to make the telescoping pattern visible, and then state why the endpoint terms simplify. In short, a neat mathematical answer is good, but an examiner awards marks for a clear mathematical *process*.

Problem 2.16: Combining binomial coefficients

It is known that

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$$

for all integers such that $1 \leq r \leq n - 1$. Do not prove this.
Find one possible set of values for p and q such that

$$\binom{2025}{86} + \binom{2025}{87} + \binom{2026}{1940} = \binom{p}{q}.$$

Solution 2.16

Using Pascal's triangle identity,

$$\binom{2025}{86} + \binom{2025}{87} = \binom{2026}{87}.$$

Also,

$$\binom{2026}{1940} = \binom{2026}{2026-1940} = \binom{2026}{86}$$

by the symmetry property of binomial coefficients.
So the given expression becomes

$$\binom{2026}{87} + \binom{2026}{86}.$$

Applying Pascal's triangle identity again,

$$\binom{2026}{86} + \binom{2026}{87} = \binom{2027}{87}.$$

Therefore one possible answer is

$$\boxed{p = 2027, \ q = 87}.$$

Takeaways 2.16

- When two binomial coefficients have the same upper index and consecutive lower indices, try Pascal's triangle immediately.
- The symmetry identity $\binom{n}{r} = \binom{n}{n-r}$ is often the key step before applying Pascal's triangle.
- In HSC multiple-step identities, simplify one pair at a time.

Problem 2.17: Equal numbers of men and women

A club has $2n$ members, with n women and n men.
A group consisting of an even number $(0, 2, 4, \ldots, 2n)$ of members is chosen, with the number of men equal to the number of women.
Show, giving reasons, that the number of ways to do this is

$$\binom{2n}{n}.$$

Solution 2.17

Suppose the group contains r women. Since the number of men must equal the number of women, the group must also contain r men.
For a fixed value of r, the number of ways to choose the group is

$$\binom{n}{r}\binom{n}{r}.$$

This is because we choose r women from the n women and r men from the n men.
Now r can take any value from 0 to n. So the total number of possible groups is

$$\sum_{r=0}^{n}\binom{n}{r}^2.$$

But from the identity

$$\binom{2n}{n} = \binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2,$$

this sum is equal to

$$\binom{2n}{n}.$$

Therefore, the number of ways to choose such a group is

$$\boxed{\binom{2n}{n}}.$$

Takeaways 2.17

- When a selection has balanced categories, fix the number chosen from one category first.
- A total count can often be written as a sum over all allowable cases.
- This problem is a direct application of the identity $\sum_{r=0}^{n}\binom{n}{r}^2 = \binom{2n}{n}$.

Problem 2.18: A binomial identity and double counting capstone

i) Use the identity
$$(1+x)^{2n} = (1+x)^n(1+x)^n$$
to show that
$$\binom{2n}{n} = \binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2,$$
where n is a positive integer.

ii) A club has $2n$ members, with n women and n men.

A group consisting of an even number $(0, 2, 4, \dots, 2n)$ of members is chosen, with the number of men equal to the number of women.

Show, giving reasons, that the number of ways to do this is
$$\binom{2n}{n}.$$

iii) From the group chosen in part (ii), one of the men and one of the women are selected as leaders.

Show, giving reasons, that the number of ways to choose the even number of people and then the leaders is
$$1^2\binom{n}{1}^2 + 2^2\binom{n}{2}^2 + \cdots + n^2\binom{n}{n}^2.$$

iv) The process is now reversed so that the leaders, one man and one woman, are chosen first. The rest of the group is then selected, still made up of an equal number of women and men.

By considering this reversed process and using part (ii), find a simple expression for the sum in part (iii).

Solution 2.18

i) Compare coefficients of x^n in

$$(1+x)^{2n} = (1+x)^n(1+x)^n.$$

The coefficient of x^n on the left is $\binom{2n}{n}$. On the right it is

$$\sum_{r=0}^{n} \binom{n}{r}\binom{n}{n-r} = \sum_{r=0}^{n} \binom{n}{r}^2.$$

Hence

$$\binom{2n}{n} = \binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2.$$

ii) If the group contains r women, it must also contain r men, so for each r there are

$$\binom{n}{r}\binom{n}{r} = \binom{n}{r}^2$$

choices. Summing over $r = 0, 1, \ldots, n$ gives

$$\sum_{r=0}^{n} \binom{n}{r}^2 = \binom{2n}{n}$$

by part (i).

iii) For a fixed r, after choosing r women and r men there are r^2 ways to choose one male leader and one female leader. So the total number of outcomes is

$$\sum_{r=1}^{n} r^2\binom{n}{r}^2 = 1^2\binom{n}{1}^2 + 2^2\binom{n}{2}^2 + \cdots + n^2\binom{n}{n}^2.$$

iv) Reverse the process. Choose the leaders first in n^2 ways. Then choose the rest of the group from the remaining $n-1$ men and $n-1$ women, with equal numbers of each. By part (ii), this can be done in

$$\binom{2n-2}{n-1}$$

ways. Therefore

$$1^2\binom{n}{1}^2 + 2^2\binom{n}{2}^2 + \cdots + n^2\binom{n}{n}^2 = \boxed{n^2\binom{2n-2}{n-1}}.$$

Takeaways 2.18

- This is a strong example of counting the same set of outcomes in two different ways.
- In part (iii), the count is grouped by the number of men and women selected.
- In part (iv), choosing the leaders first turns the same problem into a much simpler count.
- Double counting is a powerful way to derive closed forms for complicated sums.

Problem 2.19: Students and teachers in a circle

A group with 5 students and 3 teachers is to be arranged in a circle.
In how many ways can this be done if no more than 2 students can sit together?

A. $4! \times 3!$

B. $5! \times 3!$

C. $2! \times 5! \times 3!$

D. $2! \times 2! \times 2! \times 3!$

Solution 2.19

Arrange the 3 teachers in a circle first. The number of circular arrangements is

$$(3-1)! = 2!.$$

These 3 teachers create 3 gaps between them. To ensure that no more than 2 students sit together, the 5 students must be split across these 3 gaps with at most 2 students in each gap.
We need

$$x_1 + x_2 + x_3 = 5$$

with each $x_i \leq 2$.
The only possible distribution is

$$2, 2, 1.$$

There are

$$3$$

ways to choose which gap gets the single student.
Now arrange the 5 distinct students in the chosen student positions:

$$5!$$

ways.
Hence the total number of arrangements is

$$2! \times 3 \times 5! = 2! \times 3! \times 5! = 5! \times 3!.$$

So the correct answer is

$$\boxed{\text{B. } 5! \times 3!}.$$

Takeaways 2.19

- For circular arrangement problems with mixed groups, it is often best to fix one category first and then place the other category into the gaps.
- Restriction questions often become gap-distribution problems.
- After handling the pattern of slots, place the distinct people into those slots at the end.

Problem 2.20: Guests who refuse to sit together

Eight guests are to be seated at a round table. If two of these guests refuse to sit next to each other, how many seating arrangements are possible?

Solution 2.20

First count all circular seating arrangements of 8 distinct guests:

$$(8-1)! = 7! = 5040.$$

Now count the arrangements in which the two particular guests do sit next to each other.
Treat those two guests as a single block. Then we have 7 objects around the table: the block plus the other 6 guests. The number of circular arrangements is

$$(7-1)! = 6!.$$

Within the block, the two guests can switch places in

$$2!$$

ways.
So the number of arrangements in which they sit together is

$$2 \times 6! = 1440.$$

Therefore, the number of arrangements in which they do *not* sit next to each other is

$$7! - 2 \cdot 6! = 5040 - 1440 = 3600.$$

Hence the required number of seating arrangements is $\boxed{3600}$.

Takeaways 2.20

- In circular arrangement questions with a restriction like "not next to each other", complementary counting is often the cleanest method.
- To count adjacent people, bind them into one block and arrange the block with the remaining people.
- For circular permutations of n distinct objects, use $(n-1)!$.

Problem 2.21: Australian Powerball counting

In Australian Powerball, a single standard game consists of choosing 7 regular numbers from 35, and then choosing 1 Powerball from 20.

i) How many different standard Powerball game lines are possible?

ii) Hence show that the probability of winning Division 1 (all 7 regular numbers and the Powerball) is

$$\frac{1}{134490400}.$$

iii) For a fixed winning draw, in how many different game lines would a player match exactly 3 regular numbers and the Powerball? Hence show that this probability is about 1 in 188.

Solution 2.21

i) Choose 7 regular numbers from 35, then choose 1 Powerball from 20:

$$\binom{35}{7} \times 20 = 6724520 \times 20 = 134490400.$$

So there are $\boxed{134490400}$ standard game lines.

ii) For a fixed draw, exactly one line wins Division 1, so

$$P(\text{Division 1}) = \frac{1}{134490400}.$$

iii) To match exactly 3 regular numbers, choose 3 of the 7 winning regular numbers and 4 of the 28 non-winning regular numbers. The Powerball must also be correct:

$$\binom{7}{3}\binom{28}{4} = 35 \times 20475 = 716625.$$

Hence

$$P(3 + \text{Powerball}) = \frac{716625}{134490400} \approx 0.005328 \approx \frac{1}{188}.$$

So the probability is about $\boxed{1 \text{ in } 188}$.

Takeaways 2.21

- Lottery questions often split naturally into "choose the regular numbers" and "choose the Powerball".
- "Exactly k matches" means choose some winning numbers and the rest from the non-winning numbers.
- Published lottery odds are often rounded, so a combinatorial answer may be an approximation rather than a perfect exact fraction.

Problem 2.22: License plates and careful reading

A license plate involves 2 digits, followed by 2 letters, followed by 2 more digits, all chosen at random.

a) How many different license plates are possible?

b) What is the probability that the last digit is prime?

c) Find the probability that the plate says `12AB34`.

Solution 2.22

We interpret "all chosen at random" to mean repetition is allowed.

a) There are 10 choices for each digit and 26 choices for each letter:

$$10 \times 10 \times 26 \times 26 \times 10 \times 10 = 10^4 \cdot 26^2 = 6760000$$

So the number of license plates is $\boxed{6760000}$.

b) The last digit can be any of 0 to 9. The prime digits are $2, 3, 5, 7$, so there are 4 favourable digits out of 10:

$$\frac{4}{10} = \frac{2}{5}$$

Thus the probability is $\boxed{\frac{2}{5}}$.

c) There is exactly one favourable plate, namely `12AB34`, out of all possible license plates. Therefore

$$\frac{1}{6760000}.$$

$$\boxed{\frac{1}{6760000}}$$

Takeaways 2.22

- Before counting, check the format carefully.
- If a question asks for one exact plate, the probability is usually $\frac{1}{\text{total number of plates}}$.
- When repetition is not forbidden, assume repeated digits or letters are allowed.

Problem 2.23: A mixed combinatorics review

1. How many different ways can 5 letters be chosen from `URGOATED` if all 4 vowels must be included?
2. How many different ways could the letters of `COOKED` be arranged?
3. How many 4-digit numbers can be made from the digits 1 to 8 without repetition?

Explain briefly which parts are combinations and which parts are permutations.

Solution 2.23

1. This is a combination because order does not matter. We must take U, O, A, E and then choose 1 more letter from the remaining 4 consonants:

$$\binom{4}{1} = 4$$

2. This is a permutation with repeated letters:

$$\frac{6!}{2!} = 360$$

3. This is a permutation without repetition:

$$8 \times 7 \times 6 \times 5 = 1680$$

So the answers are:

$$\boxed{4,\ 360,\ 1680}$$

Takeaways 2.23

- Combinations are for selections where order does not matter.
- Permutations are for arrangements where order does matter.
- Repeated objects in a permutation require division by factorials of repeated counts.

3 Part 2: Practice and Challenge Bank

3.1 Warm-Up Drills

Problem 3.1: Expanding a binomial surd expression

Expand

$$(1-\sqrt{2})^5 = a + b\sqrt{2},$$

where a and b are integers. Find a and b.

Hint: Use the binomial theorem, then simplify powers of $\sqrt{2}$ carefully: $(\sqrt{2})^2 = 2$, $(\sqrt{2})^3 = 2\sqrt{2}$, and so on.

Solution 3.1

Using the binomial theorem,

$$\begin{aligned}(1-\sqrt{2})^5 &= 1 - 5\sqrt{2} + 10(\sqrt{2})^2 - 10(\sqrt{2})^3 + 5(\sqrt{2})^4 - (\sqrt{2})^5\\ &= 1 - 5\sqrt{2} + 20 - 20\sqrt{2} + 20 - 4\sqrt{2}\\ &= 41 - 29\sqrt{2}.\end{aligned}$$

Hence

$$\boxed{a = 41, \quad b = -29}.$$

Takeaways 3.1

- After expanding, separate the rational terms from the $\sqrt{2}$ terms.
- Powers of $\sqrt{2}$ alternate between integer multiples and surd multiples.

Problem 3.2: When is a conjugate surd sum rational?

For what integer values of n is

$$(1-\sqrt{2})^n + (1+\sqrt{2})^n$$

rational?

Hint: Expand both powers using the binomial theorem and compare the terms involving odd and even powers of $\sqrt{2}$. Also think about what happens for negative integers.

Solution 3.2

For a non-negative integer n, expand both terms by the binomial theorem:

$$(1+\sqrt{2})^n = \binom{n}{0} + \binom{n}{1}\sqrt{2} + \binom{n}{2}(\sqrt{2})^2 + \cdots,$$

$$(1-\sqrt{2})^n = \binom{n}{0} - \binom{n}{1}\sqrt{2} + \binom{n}{2}(\sqrt{2})^2 - \cdots.$$

When we add these two expansions, all terms involving odd powers of $\sqrt{2}$ cancel, while the even-power terms remain. Since

$$(\sqrt{2})^{2k} = 2^k,$$

all remaining terms are rational.
Therefore, for every non-negative integer n,

$$(1-\sqrt{2})^n + (1+\sqrt{2})^n$$

is rational.
Now consider negative integers. Since

$$(1+\sqrt{2})(1-\sqrt{2}) = 1 - 2 = -1,$$

we have

$$1-\sqrt{2} = -\frac{1}{1+\sqrt{2}}.$$

So for any positive integer n,

$$(1-\sqrt{2})^{-n} + (1+\sqrt{2})^{-n} = (-1)^n(1+\sqrt{2})^n + (-1)^n(1-\sqrt{2})^n = (-1)^n\Big((1+\sqrt{2})^n + (1-\sqrt{2})^n\Big),$$

which is also rational.
Hence the expression is rational for

$$\boxed{\text{all integers } n.}$$

Takeaways 3.2

- Conjugate surds often become much simpler when added or multiplied.
- In binomial expansions of $(a+b)^n$ and $(a-b)^n$, the odd-power terms cancel when the expansions are added.
- Here that cancellation shows the expression is rational for every integer n.

Problem 3.3: PINs with numbers and letters

The number of possible different PINs with a combination of 4 numbers and 2 letters is:

A. 4435236

B. 6760000

C. 1000000

D. 10676

Hint: Use the multiplication principle: count the choices for the 4 digits and the 2 letters separately, then multiply.

Solution 3.3

Assume the PIN has 4 digits followed by 2 letters, with repetition allowed.
There are 10 choices for each digit and 26 choices for each letter, so the total number of PINs is

$$10^4 \times 26^2 = 10000 \times 676 = 6760000.$$

So the correct answer is

$$\boxed{\text{B. } 6760000}.$$

Takeaways 3.3

- If each position can be chosen independently, multiply the number of choices for each position.
- Unless a question says otherwise, repetition is usually allowed in code or PIN questions.

Problem 3.4: Two coins of the same metal

A bag contains n metal coins, where $n \geq 3$. There are k silver coins in the bag and the rest are bronze.
Two coins are drawn at random from the bag, without replacement.
Find an expression for the probability that the two coins drawn are made of the same metal.

Hint: The two coins are the same metal if they are both silver or both bronze.

Solution 3.4

There are two favourable cases:
Both silver:

$$\frac{k}{n} \cdot \frac{k-1}{n-1}.$$

Both bronze: There are $n-k$ bronze coins, so

$$\frac{n-k}{n} \cdot \frac{n-k-1}{n-1}.$$

Hence the required probability is

$$\boxed{\frac{k(k-1)}{n(n-1)} + \frac{(n-k)(n-k-1)}{n(n-1)}}.$$

Takeaways 3.4

- For "same type" probability questions, split into mutually exclusive cases and add.
- Without replacement means the denominator changes on the second draw.

Problem 3.5: Probabilities with the four jacks

In a pack of cards, the 4 jacks are taken out and shuffled.

a) What is the probability of picking out the jack of diamonds at random?

b) If all the jacks are arranged in order, what is the probability of guessing the correct order?

Hint: For part (a), all four jacks are equally likely. For part (b), count the total number of possible orders of the four jacks.

Solution 3.5

a) There are 4 jacks, and only one of them is the jack of diamonds. So

$$P(\text{jack of diamonds}) = \frac{1}{4}.$$

b) The 4 jacks can be arranged in

$$4! = 24$$

different orders. Only one of these is the correct order, so

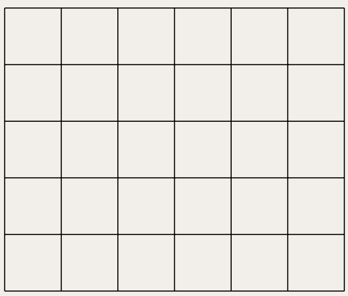
$$P(\text{guessing the correct order}) = \frac{1}{24}.$$

Takeaways 3.5

- If all outcomes are equally likely, probability is favourable outcomes over total outcomes.
- Ordered arrangements of 4 distinct objects are counted by 4!.

Problem 3.6: Counting rectangles in a 5×6 grid

How many rectangles can be found in the 5×6 grid below? Remember that squares should also be counted, since every square is a rectangle.



Hint: Choose any 2 of the vertical grid lines and any 2 of the horizontal grid lines.

Solution 3.6

A rectangle is determined by choosing:

- 2 of the 7 vertical grid lines
- 2 of the 6 horizontal grid lines.

So the total number of rectangles is

$$\binom{7}{2}\binom{6}{2} = 21 \times 15 = 315.$$

Hence the number of rectangles is $\boxed{315}$.

Takeaways 3.6

- Grid-rectangle questions are usually solved by choosing boundary lines, not by listing shapes one by one.
- Squares are already included, since every square is also a rectangle.

Problem 3.7: How fast does brute force grow?

A password is made from n binary digits, each of which is either 0 or 1.

a) How many possible passwords are there when $n = 5$?

b) How many possible passwords are there when $n = 10$?

c) Explain briefly why checking every possible password becomes difficult when n is large.

Hint: Each position has 2 choices, so use the multiplication principle.

Solution 3.7

a) For $n = 5$, the number of passwords is

$$2^5 = 32.$$

b) For $n = 10$, the number of passwords is

$$2^{10} = 1024.$$

c) The total number of possibilities doubles each time one more digit is added. So the number of passwords grows exponentially, and brute-force checking quickly becomes impractical for large n.

Takeaways 3.7

- Some counting problems grow exponentially, even when the rule looks simple.
- This is one reason efficient counting methods matter.

Problem 3.8: Repeated derivatives of $\frac{1}{x}$

If

$$f(x) = \frac{1}{x},$$

find:

i) $f'(x)$

ii) $f''(x)$

iii) $f^{(5)}(x)$

iv) $f^{(n)}(x)$.

Hint: Write

$$f(x) = x^{-1}$$

and differentiate a few times to spot the pattern.

Solution 3.8

Using the power rule,

$$f(x) = x^{-1}.$$

i)

$$f'(x) = -x^{-2} = -\frac{1}{x^2}.$$

ii)

$$f''(x) = 2x^{-3} = \frac{2}{x^3}.$$

iii) Continuing the pattern,

$$f^{(3)}(x) = -\frac{6}{x^4}, \qquad f^{(4)}(x) = \frac{24}{x^5},$$

so

$$f^{(5)}(x) = -\frac{120}{x^6}.$$

iv) Each differentiation introduces:

- one extra factor in the numerator
- one extra power of x in the denominator
- alternating signs.

Therefore

$$\boxed{f^{(n)}(x) = (-1)^n \frac{n!}{x^{n+1}}}.$$

Takeaways 3.8

- Repeated differentiation often reveals a pattern after the first few derivatives.
- Here the coefficients are factorials: $1, 1, 2, 6, 24, 120, \ldots$.
- This is more of a calculus enrichment problem than a core combinatorics problem, but it connects nicely to factorial notation.

Problem 3.9: Using Pascal's triangle to differentiate x^4

i) Use Pascal's triangle to expand

$$(x+h)^4.$$

ii) If

$$f(x) = x^4,$$

simplify

$$f(x+h) - f(x).$$

iii) Hence use the definition

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

to differentiate x^4.

iv) For the general case

$$f(x) = x^n,$$

write down the expansion of $(x+h)^n$, and hence show that

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

Hint: Use row 4 of Pascal's triangle for the specific case, then use the binomial theorem for $(x+h)^n$. In both cases, factor out h before taking the limit.

Solution 3.9

i) Using Pascal's triangle or the binomial theorem,

$$(x+h)^4 = x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4.$$

ii) Since $f(x) = x^4$,

$$f(x+h) - f(x) = (x+h)^4 - x^4 = 4x^3h + 6x^2h^2 + 4xh^3 + h^4.$$

iii) Therefore,

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\ &= \lim_{h\to 0} \frac{4x^3h + 6x^2h^2 + 4xh^3 + h^4}{h} \\ &= \lim_{h\to 0} \left(4x^3 + 6x^2h + 4xh^2 + h^3\right) \\ &= 4x^3. \end{aligned}$$

Hence

$$\boxed{f'(x) = 4x^3.}$$

iv) In general, the binomial theorem gives

$$(x+h)^n = x^n + \binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n.$$

So if

$$f(x) = x^n,$$

then

$$f(x+h) - f(x) = nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n.$$

Therefore,

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\ &= \lim_{h\to 0} \left(nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + h^{n-1}\right) \\ &= nx^{n-1}. \end{aligned}$$

Hence

$$\boxed{\frac{d}{dx}(x^n) = nx^{n-1}.}$$

Takeaways 3.9

- Pascal's triangle gives the coefficients in expansions such as $(x+h)^4$.
- In the derivative definition, the key step is to factor out h before taking the limit.
- The same idea works in general: the binomial theorem for $(x+h)^n$ leads to the power rule $\dfrac{d}{dx}(x^n) = nx^{n-1}$.
- This is a calculus enrichment problem, but it shows one way algebra and Pascal's triangle support differentiation.

3.2 Stretch Problems

Problem 3.10: Term independent of x

Find the term independent of x in the expansion of

$$\left(11x - \frac{4}{x^2}\right)^{15}.$$

Hint: Use the general term in the binomial expansion. Then make the power of x equal to 0.

Solution 3.10

The general term is

$$T_{r+1} = \binom{15}{r}(11x)^{15-r}\left(-\frac{4}{x^2}\right)^r.$$

So the power of x is

$$x^{15-r}x^{-2r} = x^{15-3r}.$$

For the term independent of x, we need

$$15 - 3r = 0 \quad \Rightarrow \quad r = 5.$$

Hence the required term is

$$\binom{15}{5}(11x)^{10}\left(-\frac{4}{x^2}\right)^5 = \binom{15}{5}\cdot 11^{10}\cdot(-4)^5$$
$$= 3003 \cdot 11^{10} \cdot (-1024).$$

Therefore, the term independent of x is

$$\boxed{-3003 \cdot 1024 \cdot 11^{10}}.$$

Takeaways 3.10

- For "term independent of x" questions, write the general term first.
- The key step is solving the exponent equation that makes the power of x equal to 0.

Problem 3.11: Five-letter words from `Mamungkukumpurangkuntjunya`

How many 5-letter "words" can be made from the word `Mamungkukumpurangkuntjunya`? *Helpful note:* To save time, you may use the repeated-letter frequencies

$$u^6,\ n^4,\ m^3,\ a^3,\ k^3,\ g^2,$$

and note that the remaining letters each occur once.

Hint: Treat this as an arrangement problem with repeated letters. Split into cases according to the repetition pattern of the 5 letters.

Solution 3.11

The word `Mamungkukumpurangkuntjunya` contains the letter types

$$a, g, j, k, m, n, p, r, t, u, y$$

with frequencies

$$u^6,\ n^4,\ m^3,\ a^3,\ k^3,\ g^2,$$

and p, r, t, j, y each occurring once.
We count by repetition pattern.

1. All 5 letters distinct:

$$\binom{11}{5}5! = 55440.$$

2. One pair and 3 singles:

$$6 \cdot \binom{10}{3} \cdot \frac{5!}{2!} = 43200.$$

3. Two pairs and 1 single:

$$\binom{6}{2} \cdot 9 \cdot \frac{5!}{2!2!} = 4050.$$

4. One triple and 2 singles:

$$5 \cdot \binom{10}{2} \cdot \frac{5!}{3!} = 4500.$$

5. One triple and one pair:

$$5 \cdot 5 \cdot \frac{5!}{3!2!} = 250.$$

6. Four of one letter and one single:

$$2 \cdot 10 \cdot \frac{5!}{4!} = 100.$$

7. Five of one letter:

$$1.$$

Therefore, the total number of 5-letter words is

$$55440 + 43200 + 4050 + 4500 + 250 + 100 + 1 = 107541.$$

Hence the answer is $\boxed{107541}$.

Takeaways 3.11

- When letters repeat, count by repetition pattern such as $(2, 1, 1, 1)$ or $(3, 2)$.
- After choosing the letters, use multinomial arrangements such as $\frac{5!}{2!}$ or $\frac{5!}{3!2!}$.

Problem 3.12: Constant term from a product of two expansions

Find the term independent of x in the expansion of

$$\left(x + \frac{1}{x}\right)^5 \left(2x^2 - \frac{3}{x}\right)^6.$$

Hint: Write the general term from each bracket, then find which pair of terms gives a total power of x equal to 0.

Solution 3.12

From
$$\left(x+\frac{1}{x}\right)^5,$$
the general term is
$$\binom{5}{r}x^{5-r}\left(\frac{1}{x}\right)^r=\binom{5}{r}x^{5-2r}.$$
From
$$\left(2x^2-\frac{3}{x}\right)^6,$$
the general term is
$$\binom{6}{s}(2x^2)^{6-s}\left(-\frac{3}{x}\right)^s=\binom{6}{s}2^{6-s}(-3)^s x^{12-3s}.$$
So in the product, the power of x is
$$(5-2r)+(12-3s)=17-2r-3s.$$
For a constant term, we need
$$17-2r-3s=0.$$
With $0\le r\le 5$ and $0\le s\le 6$, the only solution is
$$r=1,\quad s=5.$$
So the constant term is
$$\begin{aligned}\binom{5}{1}\binom{6}{5}\cdot 1\cdot 2^1(-3)^5&=5\cdot 6\cdot 2\cdot(-243)\\&=-14580.\end{aligned}$$
Hence the term independent of x is
$$\boxed{-14580}.$$

Takeaways 3.12

- In products of expansions, add the exponents from the chosen terms.
- The constant term occurs when the total exponent of x is 0.
- After finding the exponent condition, substitute the matching values back into the coefficients.

Problem 3.13: Adults and children at the cinema

Three adults and five children go to the cinema and are seated next to each other in a row of eight seats.
How many ways can these eight people sit so that at least two of the adults sit next to each other?

Hint: Count all seatings first, then subtract the seatings in which no two adults sit together.

Solution 3.13

There are

$$8!$$

ways to seat the 8 distinct people in a row.
Now count the arrangements in which no two adults sit together.
First arrange the 5 children:

$$5!$$

ways.
This creates 6 gaps where adults could be placed:

$$\_\ C\ \_\ C\ \_\ C\ \_\ C\ \_\ C\ \_$$

Choose 3 of these 6 gaps for the adults:

$$\binom{6}{3}$$

ways.
Then arrange the 3 adults in those chosen gaps:

$$3!$$

ways.
So the number of seatings with no two adults together is

$$5!\binom{6}{3}3! = 120 \cdot 20 \cdot 6 = 14400.$$

Therefore, the number of seatings in which at least two adults sit together is

$$8! - 14400 = 40320 - 14400 = 25920.$$

Hence the required number of ways is

$$\boxed{25920}.$$

Takeaways 3.13

- "At least" often suggests complementary counting.
- To keep certain people apart, arrange the others first and then place people into the gaps.
- Gap methods are especially useful for adjacency restrictions in line arrangements.

Problem 3.14: The alternating row constraint

Four boys, including Bob, and four girls, including Grace, are to stand in a row.
Find the number of distinct arrangements if the boys and girls must alternate positions and Bob must stand immediately next to Grace.

Hint: Count all alternating arrangements first. Then use symmetry: in an alternating row, each specific boy-girl pairing appears equally often.

Solution 3.14

There are two alternating patterns:

$$\text{B G B G B G B G} \qquad \text{or} \qquad \text{G B G B G B G B}.$$

For each pattern, the boys can be arranged in 4! ways and the girls in 4! ways. So the total number of alternating arrangements is

$$2 \cdot 4! \cdot 4! = 1152.$$

Now impose the condition that Bob is next to Grace.
In any alternating row of 8 people, there are 7 adjacent boy-girl pairs. On the other hand, Bob can be paired with any of the 4 girls, and Grace can be paired with any of the 4 boys, giving

$$4 \times 4 = 16$$

possible boy-girl pairings in total.
By symmetry, each specific boy-girl pairing occurs equally often among the alternating arrangements, so the fraction in which Bob and Grace are adjacent is

$$\frac{7}{16}.$$

Hence the required number of arrangements is

$$1152 \cdot \frac{7}{16} = 72 \cdot 7 = 504.$$

Therefore the number of distinct arrangements is

$$\boxed{504}.$$

Takeaways 3.14

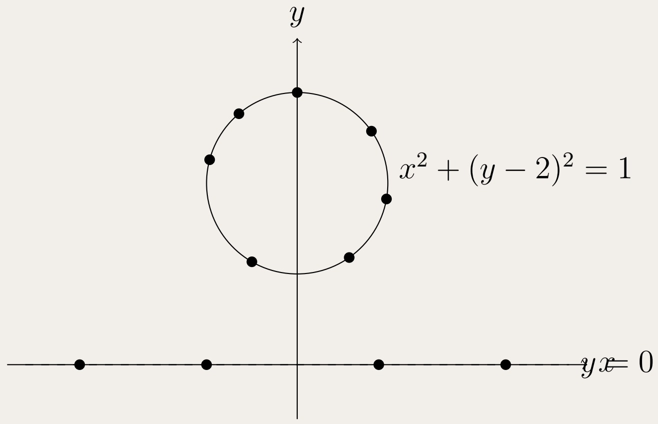
- With equal group sizes, alternating rows always begin from exactly two templates.
- Count the full structured sample space first, then apply the extra adjacency restriction.
- This problem is a good example of symmetry simplifying what would otherwise be a messy case split.
- Numbering the positions from 1 to 8. There are 7 adjacent pairs for Bob and Grace: $(1,2), (2,3), \ldots, (7,8)$.

Problem 3.15: Lines determined by points on a line and a circle

There are 11 points on the plane.

- 4 of them lie on the line $y = 0$
- 7 of them lie on the circle
$$x^2 + (y-2)^2 = 1.$$

How many different lines can be formed that pass through at least two of the 11 points?



Hint: Start with all pairs of points, then correct for the overcount caused by the 4 collinear points on $y = 0$.

Solution 3.15

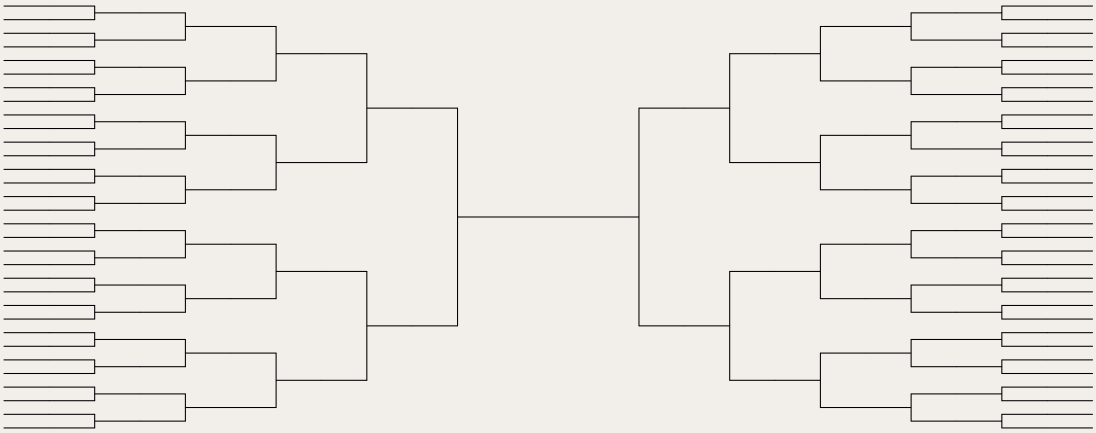
If no three points were collinear, the number of lines would be
$$\binom{11}{2} = 55.$$
However, the 4 points on the line $y = 0$ are collinear. The pairs among these 4 points are
$$\binom{4}{2} = 6,$$
but they produce only *one* line.
So among these 6 counted pairs, we have overcounted by
$$6 - 1 = 5.$$
Therefore, the number of distinct lines is
$$\binom{11}{2} - 5 = 55 - 5 = 50.$$
Hence the required number of lines is
$$\boxed{50}.$$

Takeaways 3.15

- The usual starting point is $\binom{n}{2}$, because each pair of points determines a line.
- If several points are collinear, those pairs all represent the same line and must be adjusted.
- A block of k collinear points contributes $\binom{k}{2}$ pairs but only 1 line.

Problem 3.16: Dota knockout bracket

Suppose a college Dota tournament consists of 64 teams playing head-to-head in a knockout tournament. There are 6 rounds: the round of 64, round of 32, round of 16, round of 8, the final four teams, and the grand final.
Suppose you are filling out a bracket that specifies which team will win each match in each round.



How many possible brackets can you make?

Hint: A completed bracket is determined by choosing the winner of every match. How many matches are played in a 64-team knockout tournament?

Solution 3.16

In a knockout tournament, each match eliminates exactly one team.
Starting with 64 teams, we must eliminate 63 teams to get one champion. So the tournament contains

$$63$$

matches in total.
For each match, there are 2 possible winners. Therefore the total number of possible completed brackets is

$$2^{63}.$$

Hence the number of possible brackets is

$$\boxed{2^{63}}.$$

Takeaways 3.16

- In a knockout tournament with n teams, there are $n-1$ matches.
- A bracket is determined by the winner of each match.
- Once the number of matches is known, use the multiplication principle.

Problem 3.17: Dividing a class into equal groups

A class has 28 students.
The teacher wants to divide them into 7 groups for an assignment task, with each group containing 4 students.

(i) How many different ways can this be done if the 7 groups are unlabeled?

(ii) How many different ways can this be done if the groups are labeled Group 1, Group 2, . . . , Group 7?

Hint: Start by arranging all 28 students, then correct for the fact that order within each group does not matter. In part (i), the order of the groups also does not matter, but in part (ii) it does.

Solution 3.17

If we arrange all 28 students in a line, there are

$$28!$$

ways.
Now split this line into 7 consecutive blocks of 4.
(i) If the groups are unlabeled, this overcounts in two ways:

- within each group, the 4 students can be arranged in 4! ways without changing the group
- the 7 groups themselves can be arranged in 7! ways without changing the division.

So the number of different unlabeled groupings is

$$\frac{28!}{(4!)^7\,7!}.$$

Hence

$$\boxed{\frac{28!}{(4!)^7\,7!}}.$$

(ii) If the groups are labeled, we still divide by $(4!)^7$ because order within each group does not matter, but we do *not* divide by 7! because the groups are now distinguishable.
Hence

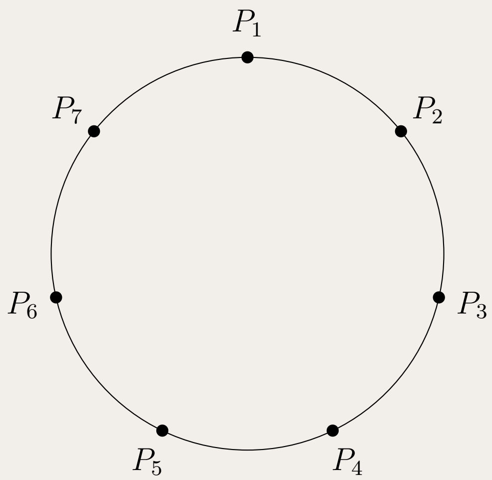
$$\boxed{\frac{28!}{(4!)^7}}.$$

Takeaways 3.17

- When dividing people into equal-sized groups, first count all arrangements, then divide by repeated internal orders.
- If the groups are unlabeled, also divide by the number of ways to reorder the groups.
- The difference between labeled and unlabeled groups here is exactly a factor of 7!.

Problem 3.18: Triangles from points on a circle

Seven points $P_1, P_2, \ldots, P_7$ are arranged in order around a circle, as shown below.



i) How many triangles can be drawn using these points as vertices?

ii) How many pairs of triangles can be drawn, where the vertices of each triangle are distinct points?

Hint: For part (i), choose any 3 of the 7 points. For part (ii), choose 6 points, then split them into two groups of 3.

Solution 3.18

i) Any 3 of the 7 points form a triangle, so the number of triangles is

$$\binom{7}{3} = 35.$$

ii) First choose the 6 points that will be used:

$$\binom{7}{6} = 7.$$

Now split those 6 points into two groups of 3 to make two triangles:

$$\binom{6}{3} = 20.$$

This counts each pair of triangles twice, because choosing Triangle A first or Triangle B first gives the same pair. So divide by 2:

$$\frac{\binom{7}{6}\binom{6}{3}}{2} = \frac{7 \cdot 20}{2} = 70.$$

Hence the answers are

$$\boxed{35 \text{ and } 70}.$$

Takeaways 3.18

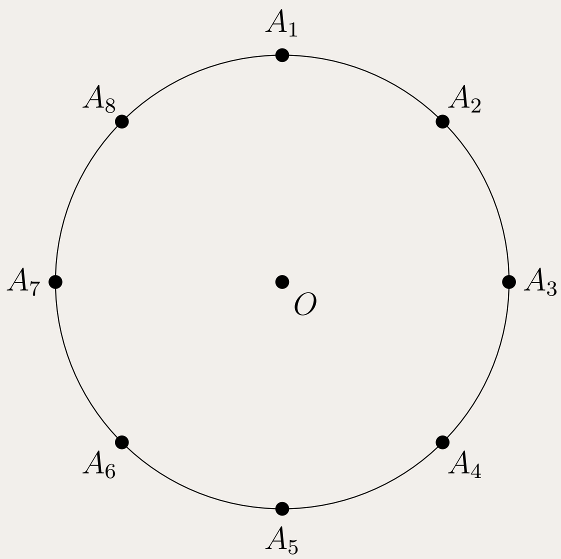
- On a circle, any choice of 3 distinct points gives a triangle.
- When counting pairs of groups, check whether the order of the groups matters.
- If the pair is unordered, divide by 2 after choosing the first and second groups.

Problem 3.19: Triangles from a regular octagon and its centre

On the unit circle with centre O, there are 8 equally spaced vertices $A_1, A_2, \ldots, A_8$. How many triangles can be formed using the 9 points

$$A_1, A_2, \ldots, A_8, O$$

as vertices?



Hint: Choose any 3 of the 9 points, then subtract the choices that are collinear.

Solution 3.19

If there were no collinear triples, the number of triangles would be

$$\binom{9}{3} = 84.$$

Now identify the collinear triples.
No line can contain three of the points on the circle, because a line meets a circle in at most two points. However, the centre O lies on each diameter, so the following four triples are collinear:

$$(A_1, O, A_5), \quad (A_2, O, A_6), \quad (A_3, O, A_7), \quad (A_4, O, A_8).$$

These 4 triples do not form triangles, so we subtract them:

$$84 - 4 = 80.$$

Hence the number of triangles is

$$\boxed{80}.$$

Takeaways 3.19

- Start with $\binom{n}{3}$ when counting triangles from points, then subtract collinear triples.
- On a circle, the only collinear triples here come from diameters passing through the centre.
- Geometry counting problems often reduce to one small structural observation.

Problem 3.20: Pigeons and pigeonholes

A flock of 41 pigeons fly into 10 different pigeonholes.

a) Show that there is at least one pigeonhole with at least 5 pigeons.

b) Show that there is at least one pair of pigeonholes in which there are at least 9 pigeons altogether.

Hint: For part (a), divide 41 by 10. For part (b), group the 10 pigeonholes into 5 pairs.

Solution 3.20

a) If every pigeonhole had at most 4 pigeons, then the total number of pigeons would be at most

$$10 \times 4 = 40.$$

But there are actually 41 pigeons, so this is impossible.
Therefore, at least one pigeonhole must contain at least

$$\boxed{5}$$

pigeons.
b) Group the 10 pigeonholes into 5 pairs.
If every pair of pigeonholes had at most 8 pigeons altogether, then the total number of pigeons would be at most

$$5 \times 8 = 40.$$

Again, this is impossible because there are 41 pigeons.
Therefore, at least one pair of pigeonholes must contain at least

$$\boxed{9}$$

pigeons altogether.

Takeaways 3.20

- The pigeonhole principle often begins with: "Suppose every group had at most this many."
- If the resulting total is too small, then some group must exceed that bound.
- For part (b), the key idea is to create larger pigeonholes by pairing the original ones.

3.3 Challenge Corner

Problem 3.21: Comparing coefficients in $(1+x)^{m+n}$

Use

$$(1+x)^{m+n} = (1+x)^m(1+x)^n$$

and compare coefficients of x^r to show that

$$\binom{m+n}{r} = \binom{m}{0}\binom{n}{r} + \binom{m}{1}\binom{n}{r-1} + \cdots + \binom{m}{r}\binom{n}{0},$$

where terms with impossible binomial coefficients are taken to be 0.

Hint: Find the coefficient of x^r on the left first. Then, on the right, think about all the ways to obtain x^r by multiplying a term from $(1+x)^m$ with a term from $(1+x)^n$.

Solution 3.21

From the binomial theorem,

$$(1+x)^{m+n} = \sum_{k=0}^{m+n} \binom{m+n}{k} x^k,$$

so the coefficient of x^r on the left is

$$\binom{m+n}{r}.$$

Also,

$$(1+x)^m = \sum_{i=0}^{m} \binom{m}{i} x^i, \qquad (1+x)^n = \sum_{j=0}^{n} \binom{n}{j} x^j.$$

To obtain an x^r term in the product, we must choose x^i from the first factor and x^{r-i} from the second. Hence the coefficient of x^r on the right is

$$\sum_{i=0}^{r} \binom{m}{i} \binom{n}{r-i}.$$

Since both expressions are the coefficient of x^r in the same polynomial,

$$\binom{m+n}{r} = \sum_{i=0}^{r} \binom{m}{i} \binom{n}{r-i}.$$

That is,

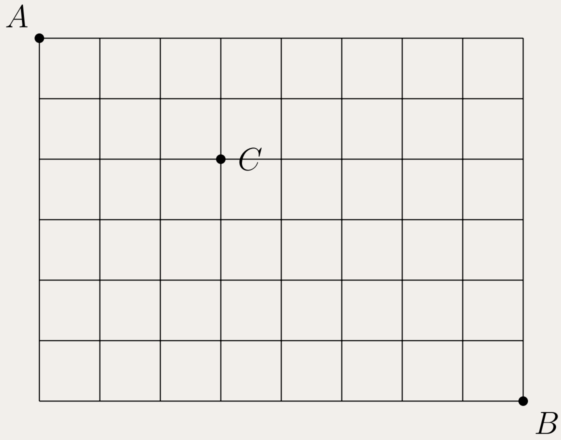
$$\boxed{\binom{m+n}{r} = \binom{m}{0}\binom{n}{r} + \binom{m}{1}\binom{n}{r-1} + \cdots + \binom{m}{r}\binom{n}{0}}.$$

Takeaways 3.21

- Expanding the same expression in two ways is a standard way to prove combinatorial identities.
- Comparing coefficients is especially useful when a problem involves powers of $(1+x)$.
- This identity is often called Vandermonde's identity.

Problem 3.22: Grid paths and binomial identities

The diagram shows a 6×8 grid. The aim is to walk from the point A in the top left-hand corner to the point B in the bottom right-hand corner by moving along the grid lines either downwards or to the right. Let C be the grid point that is 3 steps to the right and 2 steps down from A.



a) Find how many different routes are possible:

   i) without restriction

   ii) if you must pass through C.

b) Consider the diagonal of grid points

$$(0,6),(1,5),(2,4),\dots,(6,0).$$

Explain why every route from A to B passes through exactly one of these points, and hence prove that

$$\binom{14}{6} = \binom{8}{0}\binom{6}{6} + \binom{8}{1}\binom{6}{5} + \cdots + \binom{8}{6}\binom{6}{0}.$$

c) Generalise part (b) to prove that

$$\binom{m+n}{n} = \binom{m}{0}\binom{n}{n} + \binom{m}{1}\binom{n}{n-1} + \cdots + \binom{m}{n}\binom{n}{0},$$

for integers $m \ge n \ge 0$.

d) By considering an $n \times n$ square grid and the corresponding diagonal, prove that

$$\binom{2n}{n} = \binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2.$$

Hint: For part (a), count the number of right and down moves. For part (b), split each route at the unique point where it meets the diagonal. Parts (c) and (d) are the corresponding general forms of the same idea.

Solution 3.22

a)(i) A route from A to B uses 8 right moves and 6 down moves, so

$$\binom{14}{6} = \binom{14}{8}.$$

a)(ii) Split the route at C. Then

$$\#(A \to C) = \binom{5}{2} = 10, \qquad \#(C \to B) = \binom{9}{4} = 126,$$

$$\therefore\ \#(A \to C \to B) = \binom{5}{2}\binom{9}{4} = 1260.$$

b) Every move decreases $x + y$ by 1, so each route meets $x + y = 6$ exactly once, at one of

$$(0, 6), (1, 5), \dots, (6, 0).$$

$$\text{At } (k, 6-k): \quad \#(A \to (k, 6-k)) = \binom{6}{k}, \qquad \#((k, 6-k) \to B) = \binom{8}{k}.$$

$$\binom{14}{6} = \sum_{k=0}^{6} \binom{6}{k}\binom{8}{k} = \sum_{k=0}^{6} \binom{8}{k}\binom{6}{6-k}.$$

c) In an $n \times m$ grid with $m \ge n$, the total number of routes is

$$\binom{m+n}{n}.$$

Split routes by the unique point where they meet

$$x + y = n.$$

If this happens after k right moves, then

$$\#(\text{first part}) = \binom{n}{k}, \qquad \#(\text{second part}) = \binom{m}{k}.$$

$$\binom{m+n}{n} = \sum_{k=0}^{n} \binom{m}{k}\binom{n}{n-k}.$$

d) For an $n \times n$ square grid,

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}\binom{n}{n-k}.$$

Using the symmetry identity

$$\binom{n}{n-k} = \binom{n}{k},$$

we obtain

$$\boxed{\binom{2n}{n} = \binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2.}$$

Takeaways 3.22

- Grid-path counting gives a very natural combinatorial proof of Vandermonde-type identities.
- Splitting paths at a suitable diagonal point is a standard and powerful technique.
- The square-grid case leads directly to the classic identity

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2.$$

Problem 3.23: A coefficient identity for $\binom{4n}{2n}$

(i) Write down the coefficient of x^{2n} in the binomial expansion of $(1+x)^{4n}$.

(ii) Show that

$$(1+x^2+2x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^{2n-k}(x+2)^{2n-k}.$$

(iii) It is known that

$$x^{2n-k}(x+2)^{2n-k} = \binom{2n-k}{0} 2^{2n-k} x^{2n-k} + \binom{2n-k}{1} 2^{2n-k-1} x^{2n-k+1} + \cdots + \binom{2n-k}{2n-k} 2^0 x^{4n-2k}.$$

Do *not* prove this.

Show that

$$\binom{4n}{2n} = \sum_{k=0}^{n} 2^{2n-2k} \binom{2n}{k} \binom{2n-k}{k}.$$

Hint: For part (ii), rewrite $1+x^2+2x$ as $1+x(x+2)$. For part (iii), compare the coefficient of x^{2n} on both sides and identify which term in the inner expansion produces that power.

Solution 3.23

(i) From the binomial theorem,

$$(1+x)^{4n} = \sum_{r=0}^{4n} \binom{4n}{r} x^r,$$

so the coefficient of x^{2n} is $\binom{4n}{2n}$.

$$\boxed{\binom{4n}{2n}}.$$

(ii) Since $1 + x^2 + 2x = 1 + x(x+2)$,

$$\left(1 + x^2 + 2x\right)^{2n} = (1 + x(x+2))^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^{2n-k}(x+2)^{2n-k}.$$

(iii) Also,

$$(1 + x^2 + 2x)^{2n} = ((1+x)^2)^{2n} = (1+x)^{4n},$$

so its coefficient of x^{2n} is again $\binom{4n}{2n}$.
Now in

$$\sum_{k=0}^{2n} \binom{2n}{k} x^{2n-k}(x+2)^{2n-k},$$

a typical term from

$$x^{2n-k}(x+2)^{2n-k}$$

is

$$\binom{2n-k}{r} 2^{2n-k-r} x^{2n-k+r}.$$

For an x^{2n} term we need

$$2n - k + r = 2n, \qquad \text{so } r = k.$$

Hence the contribution from the kth summand is

$$\binom{2n}{k}\binom{2n-k}{k} 2^{2n-2k}.$$

This is possible only when $k \le 2n - k$, i.e. $k \le n$. Therefore

$$\binom{4n}{2n} = \sum_{k=0}^{n} 2^{2n-2k} \binom{2n}{k}\binom{2n-k}{k}.$$

Takeaways 3.23

- This is another example of proving an identity by expanding the same expression in two different ways.
- The key step is to compare coefficients of a carefully chosen power of x.
- Bounds matter: the upper limit drops from $2n$ to n because terms with $k > n$ cannot contribute to the coefficient of x^{2n}.

Problem 3.24: Repeated draws of green counters

A bag contains counters, some of which are green.
One hundred trials of an experiment are run. In each trial, one counter is selected from the bag at random and its colour noted. The counter is then returned to the bag.
Let X be the random variable representing the number of times that a green counter is selected.
Given that $E(X) = 20$, identify the distribution of X and find the probability that a green counter is selected exactly 20 times.

Hint: The experiment consists of repeated independent trials with only two outcomes on each trial: green or not green.

Solution 3.24

Since each trial is independent and each draw is with replacement, X follows a binomial distribution:

$$X \sim \mathrm{Bin}(100, p),$$

where p is the probability of selecting a green counter on any one draw.
For a binomial random variable,

$$E(X) = np.$$

Given that $E(X) = 20$, we have

$$100p = 20 \quad \Rightarrow \quad p = 0.2.$$

So

$$X \sim \mathrm{Bin}(100, 0.2).$$

Therefore,

$$P(X = 20) = \binom{100}{20}(0.2)^{20}(0.8)^{80}.$$

Hence the required probability is

$$\boxed{\binom{100}{20}(0.2)^{20}(0.8)^{80}}.$$

Takeaways 3.24

- Repeated draws with replacement lead naturally to a binomial model.
- For $X \sim \mathrm{Bin}(n, p)$, the mean is $E(X) = np$.
- Once p is known, exact probabilities are found using the binomial formula.

Problem 3.25: Arrangement formulas for x people

Find the number of arrangements possible if x people are

a) in a straight line

b) in a circle

c) in a circle with 2 particular people together

d) in a straight line with 3 particular people together

e) in a circle with 2 particular people not together.

Hint: Use $x!$ for a line and $(x-1)!$ for a circle. For "together" questions, bind people into a block. For "not together", subtract from the total.

Solution 3.25

a) In a straight line:

$$\boxed{x!}$$

b) In a circle:

$$\boxed{(x-1)!}$$

c) In a circle with 2 particular people together: Treat the 2 people as one block. Then there are $(x-1)$ objects in a circle, giving

$$(x-2)!$$

arrangements, and the pair can switch places in 2! ways. So

$$\boxed{2(x-2)!}$$

d) In a straight line with 3 particular people together: Treat the 3 people as one block. Then there are $(x-2)$ objects in a line, giving

$$(x-2)!$$

arrangements, and the 3 people inside the block can be arranged in 3! ways. So

$$\boxed{3!(x-2)! = 6(x-2)!}$$

e) In a circle with 2 particular people not together: Start with all circular arrangements:

$$(x-1)!.$$

Subtract the arrangements where the two particular people are together:

$$2(x-2)!.$$

Hence

$$\boxed{(x-1)! - 2(x-2)!}.$$

Takeaways 3.25

- The standard counts are $x!$ for a line and $(x-1)!$ for a circle.
- "Together" usually suggests treating people as a single block.
- "Not together" is often easiest by complementary counting.

Problem 3.26: A pair summing to 21

Eleven numbers are randomly chosen from the set of integers

$$S = \{1, 2, 3, 4, \ldots, 20\}.$$

Prove that the sum of two of the eleven numbers selected must equal 21.

Hint: Pair the numbers in the set so that each pair sums to 21.

Solution 3.26

The set S can be split into the following 10 pairs:

$$(1, 20),\ (2, 19),\ (3, 18),\ \ldots,\ (10, 11).$$

Each pair sums to 21.
Now suppose 11 numbers are chosen from the 20 numbers in S.
There are only 10 such pairs, but we have chosen 11 numbers. By the pigeonhole principle, at least one complete pair must be selected.
Therefore, among the 11 chosen numbers, there must be two numbers whose sum is 21.

Takeaways 3.26

- A strong first step in pigeonhole problems is to group objects into meaningful boxes.
- Here, the natural boxes are the pairs that sum to 21.
- Choosing more objects than there are boxes forces one box to contain two chosen objects.

Problem 3.27: Positive integer solutions

Show that there are

$$\binom{n-1}{k-1}$$

distinct positive integer-valued vectors $(x_1, \ldots, x_k)$ satisfying

$$x_1 + \cdots + x_k = n, \qquad x_i > 0 \text{ for all } i = 1, \ldots, k.$$

Hint: Think of writing n as a row of n identical objects, then inserting dividers to split them into k positive groups.

Solution 3.27

Represent n as n identical stars:

$$\underbrace{* * * \cdots *}_{n \text{ stars}}.$$

To split these into k positive parts, we must place $k-1$ dividers into the gaps between consecutive stars. There are

$$n-1$$

gaps between the n stars, and we choose

$$k-1$$

of them for the dividers.
Hence the number of positive integer solutions is

$$\boxed{\binom{n-1}{k-1}}.$$

Takeaways 3.27

- This is the standard stars-and-bars count for positive integer solutions.
- Positivity means every group must contain at least one star, so dividers can only go in the internal gaps.

Problem 3.28: Non-negative integer solutions

Show that there are

$$\binom{n+k-1}{k-1}$$

distinct non-negative integer-valued vectors $(x_1, \ldots, x_k)$ satisfying

$$x_1 + \cdots + x_k = n, \qquad x_i \geq 0 \text{ for all } i = 1, \ldots, k.$$

Hint: Use the same stars-and-bars idea, but now allow some groups to be empty.

Solution 3.28

Again represent n as n identical stars. This time we want to split them into k groups, but some groups may be empty.
So we place $k-1$ dividers among the n stars, allowing dividers to appear at the ends or next to each other.
This is equivalent to arranging

$$n \text{ stars and } k-1 \text{ dividers}$$

in a row. The total number of symbols is

$$n+k-1.$$

We simply choose the positions of the $k-1$ dividers:

$$\boxed{\binom{n+k-1}{k-1}}.$$

Takeaways 3.28

- This is the standard stars-and-bars count for non-negative integer solutions.
- Allowing zero means dividers may be adjacent or may appear at the ends.

Problem 3.29: Three identities from $(1+x)^n$

Consider the binomial expansion

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n.$$

i) Deduce that

$$\binom{2n}{0} + \binom{2n}{2} + \cdots + \binom{2n}{2n} = \binom{2n}{1} + \binom{2n}{3} + \cdots + \binom{2n}{2n-1}.$$

ii) Integrate both sides with respect to x from 0 to 1, and deduce that

$$\sum_{r=0}^{n} \frac{1}{r+1}\binom{n}{r} = \frac{2^{n+1}-1}{n+1}.$$

iii) Differentiate both sides with respect to x, then let $x = 1$, and deduce that

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} = n2^{n-1}.$$

Hint: For each part, start from the same binomial expansion and apply the requested operation directly to both sides.

Solution 3.29

i) Apply the identity

$$(1+x)^{2n} = \sum_{r=0}^{2n} \binom{2n}{r} x^r$$

and let $x = -1$. Then

$$\sum_{r=0}^{2n} \binom{2n}{r} (-1)^r = 0.$$

So

$$\binom{2n}{0} - \binom{2n}{1} + \binom{2n}{2} - \binom{2n}{3} + \cdots + \binom{2n}{2n} = 0.$$

Hence the sum of the even-index terms equals the sum of the odd-index terms:

$$\boxed{\binom{2n}{0} + \binom{2n}{2} + \cdots + \binom{2n}{2n} = \binom{2n}{1} + \binom{2n}{3} + \cdots + \binom{2n}{2n-1}.}$$

ii) Integrating from 0 to 1,

$$\int_0^1 (1+x)^n \, dx = \int_0^1 \sum_{r=0}^{n} \binom{n}{r} x^r \, dx.$$

The left-hand side is

$$\left[\frac{(1+x)^{n+1}}{n+1}\right]_0^1 = \frac{2^{n+1}-1}{n+1}.$$

The right-hand side is

$$\sum_{r=0}^{n} \binom{n}{r} \int_0^1 x^r \, dx = \sum_{r=0}^{n} \binom{n}{r} \frac{1}{r+1}.$$

Hence

$$\boxed{\sum_{r=0}^{n} \frac{1}{r+1} \binom{n}{r} = \frac{2^{n+1}-1}{n+1}.}$$

iii) Differentiating,

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + \cdots + n\binom{n}{n}x^{n-1}.$$

Now let $x = 1$:

$$n2^{n-1} = \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n}.$$

So

$$\boxed{\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} = n2^{n-1}.}$$

Takeaways 3.29

- A single binomial expansion can produce several useful identities by substitution, integration, and differentiation.
- Setting $x = -1$ often produces alternating-sign identities.
- Integrating or differentiating term-by-term can turn coefficients into new weighted sums.

Problem 3.30: Making an amount of money with Australian denominations

Suppose we use the simplified Australian denominations (coins less than \$1 are not used):

$$\$1,\quad \$2,\quad \$5,\quad \$10,\quad \$20,\quad \$50,\quad \$100.$$

For a fixed positive integer amount M, let $N(M)$ be the number of ways to express $\$M$ using these coins and notes, where order does *not* matter.

i) Write down a generating function whose coefficient of x^M is $N(M)$.

ii) Hence explain how the problem of finding $N(M)$ can be turned into a coefficient problem.

Enrichment note: This problem is outside the HSC Mathematics Extension 1 syllabus, but it is a very useful example of how generating functions turn a counting problem into algebra.

Hint: For each denomination d, think about the possible contributions $0, d, 2d, 3d, \ldots$.

Solution 3.30

For each denomination, we write a generating function for how many of that denomination may be used:

$$1 + x + x^2 + x^3 + \cdots \qquad \text{for } \$1,$$

$$1 + x^2 + x^4 + x^6 + \cdots \qquad \text{for } \$2,$$

and similarly for the other notes.
So the full generating function is

$$G(x) = (1 + x + x^2 + \cdots)(1 + x^2 + x^4 + \cdots)(1 + x^5 + x^{10} + \cdots)(1 + x^{10} + x^{20} + \cdots)$$
$$(1 + x^{20} + x^{40} + \cdots)(1 + x^{50} + x^{100} + \cdots)(1 + x^{100} + x^{200} + \cdots).$$

The coefficient of x^M in $G(x)$ is exactly the number of ways to make $\$M$, because each chosen power of x records how much each denomination contributes, and multiplying the choices combines them.
So

$$N(M) = [x^M]\, G(x),$$

where $[x^M]\, G(x)$ means "the coefficient of x^M in $G(x)$."
Outline of method. To solve a specific case, such as $M = 20$ or $M = 50$, expand only as far as needed and collect the coefficient of x^M. In practice, this is often done using recurrence methods or computer algebra, because the coefficients grow quickly.

Takeaways 3.30

- A generating function is a power series in which the coefficient of x^M stores the number of ways to make the amount M.
- For a sequence of numbers $a_0, a_1, a_2, \ldots$, the generating function is

$$a_0 + a_1 x + a_2 x^2 + \cdots .$$

- For the \$1 denomination, the generating function is

$$1 + x + x^2 + x^3 + \cdots$$

  because we may use $0, 1, 2, 3, \ldots$ one-dollar coins, contributing $\$0, \$1, \$2, \$3, \ldots$ respectively.
- The full generating function is the product of the generating functions for each denomination, because we can choose how many of each denomination to use independently.
- Each other denomination contributes a similar series, and multiplying these series combines the choices from all denominations.
- This is a standard higher-level combinatorics technique and is excellent enrichment beyond the HSC course.

Remark 3.1 (Algorithmic note)

We can also compute $N(M)$ using dynamic programming. This is the algorithmic version of finding the coefficient of x^M in the generating function.

```python
def calculate_coin_combinations(target_amount, denominations):
    """
    Calculates the number of ways to make a target amount using given denominations.
    This is the algorithmic equivalent of finding the coefficient of x^M
    in our generating function.
    """
    # dp is an array where dp[i] will store the number of ways to make amount i.
    # We initialize an array of zeros from 0 up to our target_amount.
    dp = [0] * (target_amount + 1)

    # Base case: There is exactly 1 way to make $0 (by choosing no coins)
    dp[0] = 1

    # Outer loop: We process one denomination bracket at a time.
    # This is exactly like multiplying the next bracket in our generating function:
    # (1 + x^2 + x^4...) * (1 + x^5 + x^10...)
```

```python
    for coin in denominations:

        # Inner loop: We update the combinations for every amount
        # from 'coin' up to the target.
        for amount in range(coin, target_amount + 1):

            # The number of ways to make the current 'amount'
              increases by the
            # number of ways we could make the amount WITHOUT this
              current coin.
            dp[amount] += dp[amount - coin]

    # The final answer is stored at the index of our target amount
    return dp[target_amount]


# --- Example Usage ---
australian_denominations = [1, 2, 5, 10, 20, 50, 100]
M = 10

ways = calculate_coin_combinations(M, australian_denominations)
print(f"To make ${M}, there are {ways} combinations.")
# Output: To make $10, there are 47 combinations.
```

Problem 3.31: Unlocking a phone by trial and error

To unlock a phone, we need to enter a 4-digit PIN.
We know that the four digits are $1, 2, 3, 4$, used distinctly, but we do not know the correct order.
If we enter a wrong PIN:

- after the first wrong attempt, the phone locks the next input for 4 seconds
- after the second wrong attempt, the phone locks the next input for 4×2 seconds
- after the third wrong attempt, the phone locks the next input for $4 \times 2 \times 2$ seconds
- and so on.

Find the average time needed to unlock the phone by trial and error, assuming each possible PIN order is equally likely to be the correct one and the time to type each PIN is negligible.

Hint: How many possible PINs are there? If the correct PIN is found on the mth attempt, how much total waiting time has occurred before that attempt?

Solution 3.31

The digits $1, 2, 3, 4$ can be arranged in

$$4! = 24$$

different ways, so if we test PINs one by one, the correct PIN is equally likely to appear in positions

$$1, 2, 3, \ldots, 24.$$

If the correct PIN is found on the mth attempt, then there have been $(m-1)$ wrong attempts before success.

The waiting times after the wrong attempts are

$$4,\ 8,\ 16,\ \ldots,\ 4 \cdot 2^{m-2}.$$

So the total waiting time before the mth attempt is

$$4 + 8 + 16 + \cdots + 4 \cdot 2^{m-2} = 4(2^{m-1} - 1).$$

Therefore the average waiting time is

$$\frac{1}{24} \sum_{m=1}^{24} 4(2^{m-1} - 1).$$

Now

$$\sum_{m=1}^{24} 2^{m-1} = 2^{24} - 1,$$

so

$$\begin{aligned} \frac{1}{24} \sum_{m=1}^{24} 4(2^{m-1} - 1) &= \frac{4}{24} \left((2^{24} - 1) - 24\right) \\ &= \frac{1}{6}(2^{24} - 25) \\ &= \frac{16777191}{6}. \end{aligned}$$

Hence the average time needed is

$$\boxed{\frac{16777191}{6} \text{ seconds}}$$

which is about

$$\boxed{2796198.5 \text{ seconds}}.$$

Takeaways 3.31

- When all possible arrangements are equally likely, the correct one is equally likely to occur in any position of your trial order.
- Exponentially growing waiting times often lead to a geometric series.
- Expected-value questions can often be solved by averaging over all possible positions of success.

Remark 3.2

"Space is big. Really big. You just won't believe how vastly, hugely, mind-bogglingly big it is. I mean, you may think it's a long way down the road to the chemist, but that's just peanuts to space."

Douglas Adams

Problem 3.32: Polynomial derivatives and a power-series expansion

Show that if

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

is any polynomial, then

$$f^{(k)}(0) = \begin{cases} a_k\, k!, & \text{for } k = 0, 1, 2, \ldots, n, \\ 0, & \text{for } k > n. \end{cases}$$

Hence explain why

$$f(x) = \frac{f(0)}{0!} + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \cdots .$$

Enrichment note: This problem is outside the HSC Mathematics Extension 1 syllabus, but it is a very useful first step toward power series and Taylor series.

Hint: Differentiate a general term $a_r x^r$ exactly k times, then substitute $x = 0$. Most terms vanish.

Solution 3.32

Write

$$f(x) = \sum_{r=0}^{n} a_r x^r.$$

If we differentiate k times, then

$$\frac{d^k}{dx^k}(a_r x^r) = \begin{cases} a_r\, r(r-1)\cdots(r-k+1)x^{r-k}, & r \geq k, \\ 0, & r < k. \end{cases}$$

Now substitute $x = 0$. Every term with $r > k$ still contains a factor of x and becomes 0. The only surviving term is the one with $r = k$, giving

$$f^{(k)}(0) = a_k\, k!$$

for $k = 0, 1, 2, \ldots, n$.
If $k > n$, every term has already vanished after enough differentiations, so

$$f^{(k)}(0) = 0.$$

Therefore

$$a_k = \frac{f^{(k)}(0)}{k!},$$

so the polynomial can be rewritten as

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n = \frac{f(0)}{0!} + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n.$$

Since f is a polynomial, this series has only finitely many nonzero terms.

Takeaways 3.32

- Repeated derivatives at $x = 0$ recover the coefficients of a polynomial.
- The factor of $k!$ appears because differentiating x^k exactly k times gives $k!$.
- For any function $f(x)$ that can be expressed as a power series, the same idea gives the Taylor series at 0:

$$f(x) = \frac{f(0)}{0!} + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots .$$

- This is the polynomial version of the idea behind Taylor series, so it is a natural enrichment topic beyond the HSC course.

Problem 3.33: Binary strings and a binomial identity

Computing is based on binary strings, which are sequences of 0s and 1s, such as 00101 and 1001110.
The problem in this question is to find the number N of binary strings that contain exactly a zeros and at most b ones, where a and b are whole numbers. We will count such strings in two different ways, and hence prove an identity.

a) Using ${}^nC_r$ notation, find how many binary strings with exactly a zeros contain:

  i) no 1s
  ii) one 1
  iii) two 1s
  iv) three 1s
  v) r 1s.

b) Hence prove that

$$N = \binom{a}{0} + \binom{a+1}{1} + \binom{a+2}{2} + \cdots + \binom{a+b}{b}.$$

c) Let S be a binary string consisting of a zeros and r ones, where $r \le b$. Extend S to a longer binary string $S01\cdots1$ by adding 0 on the right, and then adding $b-r$ ones. How many 0s and 1s are there in the resulting string, and what is its total length?

d) Describe the inverse process by which a string of $(a+1)$ zeros and b ones can be converted to a string of a zeros and at most b ones, and show that the two processes are one-to-one correspondences.

e) Hence prove that

$$N = \binom{a+b+1}{b}.$$

f) Deduce that

$$\binom{a}{0} + \binom{a+1}{1} + \binom{a+2}{2} + \cdots + \binom{a+b}{b} = \binom{a+b+1}{b}.$$

Hint: For part (a), if a string has exactly a zeros and r ones, then its length is $a+r$, and you only need to choose where the r ones go. For parts (c) and (d), build a bijection between the two families of strings.

Solution 3.33

a) If a binary string has exactly a zeros and exactly r ones, then its total length is

$$a + r.$$

We may choose the positions of the r ones (or equivalently the a zeros), so the number of such strings is

$$\binom{a+r}{r}.$$

Hence:

$$\text{i)}\ \binom{a}{0}, \qquad \text{ii)}\ \binom{a+1}{1}, \qquad \text{iii)}\ \binom{a+2}{2}, \qquad \text{iv)}\ \binom{a+3}{3}, \qquad \text{v)}\ \binom{a+r}{r}.$$

b) Since the number of ones can be $0, 1, 2, \dots, b$, summing these cases gives

$$N = \binom{a}{0} + \binom{a+1}{1} + \binom{a+2}{2} + \cdots + \binom{a+b}{b}.$$

c) Start with a binary string S containing a zeros and r ones, where $r \leq b$. Append one extra 0, then append $(b - r)$ ones. The resulting string has

$$a + 1$$

zeros and

$$r + (b - r) = b$$

ones, so its total length is

$$a + b + 1.$$

d) Conversely, given a binary string with $(a + 1)$ zeros and b ones, locate the last zero in the string and delete it together with all the ones to its right.
What remains is a binary string with exactly a zeros and at most b ones.
This process is the inverse of the construction in part (c), so the two sets of strings are in one-to-one correspondence.
e) Therefore, the number N is equal to the number of binary strings with $(a+1)$ zeros and b ones. Such strings have length

$$a + b + 1,$$

and we choose the positions of the b ones, so

$$N = \binom{a+b+1}{b}.$$

f) Equating the two expressions for N gives

$$\boxed{\binom{a}{0} + \binom{a+1}{1} + \binom{a+2}{2} + \cdots + \binom{a+b}{b} = \binom{a+b+1}{b}.}$$

Takeaways 3.33

- This is a nice example of proving an identity by counting the same set in two different ways.
- Strings with fixed numbers of zeros and ones are counted by choosing positions.
- Bijections are powerful: if two families of objects can be matched one-to-one, then they have the same size.

Problem 3.34: Derangements

A *derangement* of n distinct letters is a permutation of them so that no letter appears in its original position. For example, DABC is a derangement of ABCD, but DACB is not. Denote the number of derangements of n letters by $D(n)$.

a) By listing all the derangements of A, AB, ABC and ABCD, find the values of $D(1)$, $D(2)$, $D(3)$ and $D(4)$.

b) Suppose that we have formed a derangement of the five letters ABCDE. Let the last letter in the derangement be X, and exchange X with E. This puts E back into its original position. Either X is now also in its original position so that three letters are away from their original positions, or X is not in its original position so that four letters are away from their original positions. Hence explain why

$$D(5) = 4D(4) + 4D(3).$$

c) Use this formula to evaluate $D(5)$. Then apply the corresponding arguments and formulae to evaluate $D(6)$, $D(7)$ and $D(8)$.

d) Explain how this can be generalised to

$$D(n) = n!\left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}\right).$$

e) The alternating sum in part (d) approaches $\frac{1}{e}$ as $n \to \infty$. Explain what this means for the ratio

$$\frac{D(n)}{n!}$$

and give both a combinatorial and a probabilistic interpretation of this result.

Enrichment note: Derangements are a classic higher-level counting topic. They are not part of the standard HSC Mathematics Extension 1 syllabus, but they are an excellent example of recurrence relations in combinatorics.

Hint: For part (a), list small cases directly. For parts (b) and (c), fix where E goes, then split into the two cases described in the question. For part (d), think about inclusion-exclusion: start from all permutations and subtract those with one or more fixed points.

Solution 3.34

a) By direct listing:

$$D(1) = 0, \qquad D(2) = 1, \qquad D(3) = 2, \qquad D(4) = 9.$$

b) In a derangement of ABCDE, the letter E cannot stay in the last position, so it has 4 possible positions. Let the last letter be X, and swap E with X. Either X lands in its own position, giving $D(3)$ possibilities, or it does not, giving $D(4)$ possibilities. Hence

$$D(5) = 4D(4) + 4D(3).$$

c) Using $D(4) = 9$ and $D(3) = 2$,

$$D(5) = 4 \cdot 9 + 4 \cdot 2 = 44.$$

More generally,

$$D(n) = (n-1)\big(D(n-1) + D(n-2)\big).$$

Thus

$$\boxed{D(5) = 44, \quad D(6) = 265, \quad D(7) = 1854, \quad D(8) = 14833.}$$

d) Let A_i be the set of permutations in which the ith letter stays fixed. Then

$$|A_i| = (n-1)!, \qquad |A_i \cap A_j| = (n-2)!,$$

and in general fixing k specified letters leaves

$$(n-k)!$$

permutations. Hence, by inclusion-exclusion,

$$D(n) = n! - \binom{n}{1}(n-1)! + \binom{n}{2}(n-2)! - \cdots + (-1)^n \binom{n}{n}(n-n)!.$$

Using $\binom{n}{k}(n-k)! = \dfrac{n!}{k!}$, we can conclude that,

$$\boxed{D(n) = n!\left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}\right).}$$

e) Dividing by $n!$ gives

$$\frac{D(n)}{n!} = \frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}.$$

As $n \to \infty$, the right-hand side approaches $\dfrac{1}{e}$, so

$$\boxed{\frac{D(n)}{n!} \to \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} = \frac{1}{e}.}$$

Takeaways 3.34

- *Combinatorially,* about a fraction $\frac{1}{e}$ of all permutations are derangements for large n.
- *Probabilistically,* a random permutation has no fixed points with probability approaching $\frac{1}{e}$.
- A derangement is a permutation with no fixed points.
- Small cases are often best handled by direct listing before a general pattern appears.
- This problem is a good example of a recurrence relation arising from careful casework.
- Inclusion-exclusion gives a closed formula for $D(n)$ by counting permutations with fixed points.
- The ratio $\frac{D(n)}{n!}$ approaching $\frac{1}{e}$ means derangements make up about $\frac{1}{e}$ of all permutations when n is large.
- The Taylor series for e^x is

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

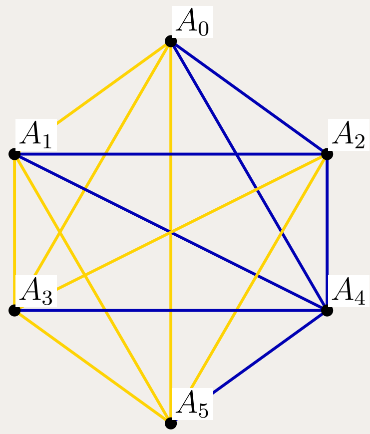
Problem 3.35: A monochromatic triangle on six vertices

A regular octahedron has 6 vertices. Label them $A_0, A_1, \ldots, A_5$. Each pair of vertices is joined by a rod, and each rod is coloured either yellow or blue.

a) Explain why there are 15 rods.

b) Each set of three vertices, together with the rods joining them, forms a triangle. Explain why there are 20 such triangles.

c) Explain why there must be at least one triangle whose rods all have the same colour.



Enrichment note: This is a small and beautiful example of a Ramsey-type idea. It is beyond the standard HSC Mathematics Extension 1 syllabus, but it is a great illustration of how counting and the pigeonhole principle can force structure.

Hint: For part (c), focus on one vertex, say A_0, and look at the 5 rods from that vertex. Use the pigeonhole principle on their colours.

Solution 3.35

a) A rod joins a pair of distinct vertices. Since there are 6 vertices, the number of rods is

$$\binom{6}{2} = 15.$$

b) Any choice of 3 vertices determines a triangle. So the number of triangles is

$$\binom{6}{3} = 20.$$

c) Focus on the vertex A_0. It is joined to the other 5 vertices by 5 rods, and each rod is coloured either yellow or blue.
By the pigeonhole principle, at least 3 of these 5 rods must have the same colour. Without loss of generality, suppose

$$A_0A_1,\ A_0A_2,\ A_0A_3$$

are all yellow.
Now consider the triangle formed by A_1, A_2, A_3.

- If one of the rods A_1A_2, A_1A_3, or A_2A_3 is yellow, then together with A_0 it forms a yellow triangle.
- If none of those rods is yellow, then all three are blue, so $A_1A_2A_3$ is a blue triangle.

In either case, there is at least one triangle whose rods all have the same colour.

Takeaways 3.35

- This is a classic example of combining counting with the pigeonhole principle.
- Looking at all edges from one vertex is the key step that forces a repeated colour.
- This is the first nontrivial case of a famous Ramsey-theory fact: in any two-colouring of the edges of K_6, there is a monochromatic triangle.

Problem 3.36: A more general form of the binomial theorem

The classical binomial theorem expands

$$(1+x)^n$$

when n is a non-negative integer. In more advanced mathematics, a similar expansion can be used when n is negative, fractional, or any real number, provided we interpret the right-hand side as an infinite power series.

a) Show that for a non-negative integer n, the binomial expansion can be written as

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots .$$

b) Assuming the same pattern continues for more general values of n, generate the power-series expansions of:

i) $\dfrac{1}{1-x}$

ii) $\dfrac{1}{(1-x)^2}$

iii) $\dfrac{1}{(1+x)^2}$

iv) $\sqrt{1+x}$.

c) Verify, using your expansions in parts (b)(i) and (b)(ii), that

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{1}{(1-x)^2},$$

assuming that the power series may be differentiated term-by-term.

Enrichment note: This problem is outside the HSC Mathematics Extension 1 syllabus. It connects the classical binomial theorem to infinite power series, Taylor series, and later to Gamma-function ideas that allow binomial coefficients to be extended beyond whole numbers.

Hint: In part (a), rewrite the usual binomial coefficients in factorial form. In part (b), substitute values such as $n=-1$, $n=-2$, and $n=\frac{1}{2}$ into the general pattern.

Solution 3.36

a) For a non-negative integer n, the usual binomial theorem gives

$$(1+x)^n = \sum_{r=0}^{n} \binom{n}{r} x^r.$$

Now

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!},$$

so the expansion may be written as

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots.$$

b)(i) Take $n = -1$:

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \cdots.$$

So

$$\boxed{\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots.}$$

b)(ii) Take $n = -2$:

$$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \cdots.$$

So

$$\boxed{\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + 4x^3 + \cdots.}$$

b)(iii) Replace x by $-x$ in part (ii):

$$\boxed{\frac{1}{(1+x)^2} = 1 - 2x + 3x^2 - 4x^3 + \cdots.}$$

b)(iv) Take $n = \frac{1}{2}$:

$$(1+x)^{1/2} = 1 + \frac{1}{2}x + \frac{\frac{1}{2}(-\frac{1}{2})}{2!}x^2 + \frac{\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})}{3!}x^3 + \cdots.$$

So

$$\boxed{\sqrt{1+x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \frac{5}{128}x^4 + \cdots}.$$

c) From part (b)(i),

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots.$$

Differentiating term-by-term gives

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = 1 + 2x + 3x^2 + 4x^3 + \cdots.$$

But this is exactly the expansion found in part (b)(ii), so

$$\boxed{\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{1}{(1-x)^2}.}$$

Takeaways 3.36

- The usual finite binomial theorem is the beginning of a much more general power-series expansion.
- For any real number n, and for $|x| < 1$,

$$(1+x)^n = \sum_{r=0}^{\infty} \binom{n}{r} x^r = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots .$$

- When n is not a non-negative integer, the binomial expansion usually becomes an infinite Taylor-style series rather than a finite polynomial. For any real value of n,

$$(1+x)^n = \sum_{r=0}^{\infty} \frac{f^{(r)}(0)}{r!}x^r, \qquad \text{where } f(x) = (1+x)^n.$$

  Refer to the problem "Polynomial derivatives and a power-series expansion" for the idea of finding these coefficients using derivatives at 0.

- The generalized coefficient may be written using Gamma functions:

$$\binom{n}{r} = \frac{\Gamma(n+1)}{\Gamma(r+1)\Gamma(n-r+1)},$$

  which extends the factorial formula to non-integer values of n.

- This topic is outside HSC Mathematics Extension 1, but it is excellent enrichment for students interested in university mathematics.

Problem 3.37: Isomers — a possible computing project

Because of branching, it is difficult to count the number of possible hydrocarbons with, say, 20 carbon atoms.
For example, for butane C_4H_{10} there are two carbon chains, and for pentane C_5H_{12} there are three carbon chains.
Investigate the number of possible carbon chains for hydrocarbons with increasing numbers of carbon atoms.

a) Explain why this is a counting problem about *connected graphs* with no cycles, where each carbon atom has degree at most 4.

b) Find the numbers of possible carbon chains for $n = 1, 2, 3, 4, 5, 6, 7$ carbon atoms.

c) If you can write code, design a program to generate all possible carbon chains for a given n and remove duplicates up to relabelling.

Enrichment note: This is a hard problem. Writing code is strongly recommended. It is outside the HSC Mathematics Extension 1 syllabus, but it is an excellent project linking combinatorics, graph theory, chemistry, and computing.

Hint: Model each carbon chain as a tree: vertices are carbon atoms and edges are carbon-carbon bonds. Two drawings represent the same chain if they are the same up to relabelling of the vertices.

Solution 3.37

a) A carbon chain can be modelled by a graph in which:

- each vertex represents a carbon atom
- each edge represents a carbon-carbon bond.

For an alkane chain, the carbon skeleton is connected and has no cycles, so it is a tree. Also, each carbon atom can make at most 4 bonds, so each vertex has degree at most 4.

b) The numbers of possible carbon chains for small n are:

n	1	2	3	4	5	6	7
number of chains	1	1	1	2	3	5	9

These are the counts of non-isomorphic trees on n vertices with maximum degree at most 4. For small n, the degree restriction does not yet exclude any tree.

c) A practical computational approach is:

- generate trees on n vertices by adding one new vertex at a time
- reject any graph with a vertex of degree more than 4
- identify graphs that are the same up to relabelling by converting them to a canonical form
- count the distinct canonical forms.

Remark 3.3 (Python idea)

The following Python snippet generates non-isomorphic trees on n vertices by adding one new vertex at a time and keeping one canonical representative of each isomorphism class. It is suitable for small values such as $n \leq 7$.

```
from itertools import permutations


def canonical_form(adj):
    n = len(adj)
    best = None
    for perm in permutations(range(n)):
        bits = []
        for i in range(n):
            for j in range(i + 1, n):
                bits.append("1" if perm[j] in adj[perm[i]] else "0")
        code = "".join(bits)
        if best is None or code < best:
            best = code
    return best


def generate_trees(n):
    if n == 1:
        return [tuple([frozenset()])]

```



```
    previous = generate_trees(n - 1)
    seen = {}

    for tree in previous:
        base = [set(neighbours) for neighbours in tree]
        for v in range(n - 1):
            if len(base[v]) >= 4:
                continue

            graph = [set(neighbours) for neighbours in base] + [set()]
            graph[v].add(n - 1)
            graph[n - 1].add(v)

            key = canonical_form(graph)
            seen[key] = tuple(frozenset(x) for x in graph)

    return list(seen.values())


for n in range(1, 8):
    print(n, len(generate_trees(n)))
```

Takeaways 3.37

- This problem is better viewed as a graph-theory and computing project than as a routine exam question.
- **Combinatorial explosion** means the number of possible structures grows extremely quickly as n increases, so direct listing soon becomes impractical.
- Because of this explosion, writing code is often the most realistic way to investigate the problem.
- This is outside the HSC Mathematics Extension 1 syllabus, but it is an excellent example of how combinatorics connects to chemistry and computer science.

4 Conclusion

Combinatorics rewards careful reading as much as technical skill. Many HSC questions become much easier once you decide whether the task is an arrangement, a selection, or a probability question built from counting. As this booklet grows, it can become a compact reference for the most common combinatorics patterns students meet in Extension 1.

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A Appendices

A.1 Gamma and Beta Functions

Although this booklet focuses on HSC Mathematics Extension 1, it is worth knowing that factorials and combinations have important extensions in higher mathematics.

Gamma function. The Gamma function is defined for suitable real or complex inputs by

$$\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}\,dt.$$

It satisfies the recurrence

$$\Gamma(z+1) = z\Gamma(z),$$

which mirrors the factorial rule $n! = n(n-1)!$. In fact,

$$\Gamma(n) = (n-1)!, \qquad \text{so} \qquad n! = \Gamma(n+1).$$

This means the Gamma function is the natural continuous extension of the factorial function.

Beta function. The Beta function is closely related to Gamma and is defined by

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

Since combinations are built from factorials,

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{\Gamma(n+1)}{\Gamma(r+1)\Gamma(n-r+1)}.$$

So Gamma and Beta functions provide a continuous version of the factorial and binomial-coefficient ideas that appear throughout combinatorics.

Why this matters. These functions form a bridge between discrete counting and continuous mathematics. They are useful in advanced probability, statistics, asymptotics, and areas such as machine learning.

A.2 Stars and Bars

Stars and bars is a method for counting integer solutions to equations such as

$$x_1 + x_2 + \cdots + x_k = n.$$

The idea is to represent the total n by n stars and use $k-1$ bars to divide the stars into k groups.

$$* \ * \Big| * \ * \ * \Big| * \ *$$

$$2 \mid 3 \mid 2$$

If the variables are required to be positive, then each group must contain at least one star, so the number of solutions is

$$\binom{n-1}{k-1}.$$

If the variables are allowed to be zero, then empty groups are allowed, so the number of solutions is

$$\binom{n+k-1}{k-1}.$$

Stars and bars is a standard combinatorial technique for turning algebraic equations into counting problems. It is directly connected to the two Part 2 challenge problems on positive and non-negative integer solutions.

Exam note. Recent HSC Mathematics Extension 1 exams do not usually ask this type of problem directly. However, it is still very good to know, because it is a powerful counting technique to keep as a mathematical "weapon" for olympiad-style problems, extension work, and future university study.

A.3 Algorithmic Counting and Complexity

Algorithmic counting is the study of methods for counting objects systematically. Complexity asks how difficult a counting problem becomes as the size of the problem grows.

Some counting problems have simple closed formulas, such as

$$\binom{n}{r}, \qquad n!, \qquad 2^n.$$

However, many counting problems grow extremely quickly and cannot be solved efficiently by checking every possibility one by one.

For example:

n	$n!$	2^n
5	120	32
10	3628800	1024
20	2432902008176640000	1048576

So even a simple arrangement problem can become enormous very quickly.

This is one reason why combinatorics is important: good counting methods, symmetry arguments, recurrences, and identities help us avoid brute force.

Syllabus note. This topic is outside the HSC Mathematics Extension 1 syllabus. However, it is very useful for higher education, especially in areas such as computer science, discrete mathematics, and algorithm design.

A.4 The Inclusion-Exclusion Principle

The inclusion-exclusion principle is used when counting objects that may belong to overlapping sets.

For two sets A and B,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

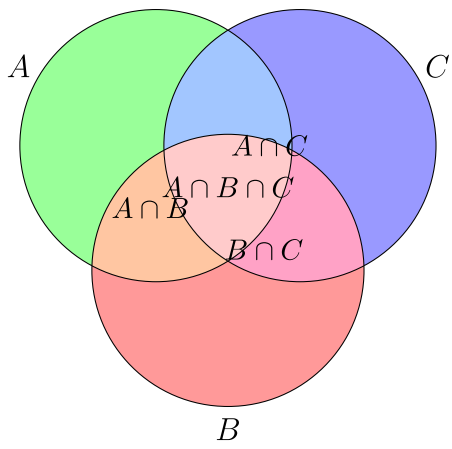
For three sets A, B, and C,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

The idea is simple: when we add the sizes of sets, elements in overlaps are counted more than once, so we must correct for this.

Also note that this is an important topic when you learn probability.

This principle is very useful in restricted counting problems, probability, and Venn diagram questions.



A.5 Concepts: Injective, Surjective, and Bijective

Understanding these concepts is incredibly important for combinatorics, particularly when answering "how many" or proving that two sets have the same number of elements without actually counting them one by one. In combinatorics, formalizing a counting process as a function provides powerful insights:

- **Injective (One-to-one) tracking:** If you can construct an injective mapping from set X to set Y, you instantly prove that $|X| \le |Y|$ (the size of X is less than or equal to Y). If you try to map more elements into fewer targets, an injection is impossible—this is the formal basis of the **Pigeonhole Principle**. When counting arrangements, injections describe ways to assign objects to distinct slots without repetition (like ${}^nP_r$).

- **Surjective (Onto) coverage:** If you can construct a surjective mapping from set X to set Y, you prove that $|X| \ge |Y|$. In counting, surjections are used when we want to count the number of ways to distribute objects into boxes where *no box is left empty*.

- **Bijective (Perfect pairing) counting:** If you can construct a bijection between set X and set Y, you prove that $|X| = |Y|$. This is called **Bijective Proof** and it is one of the most elegant techniques in combinatorics. If set X is very hard to count, but you can link it perfectly to set Y which is easy to count, then you have solved the problem! (For instance, stars and bars is a bijection between dividing items and placing dividers).

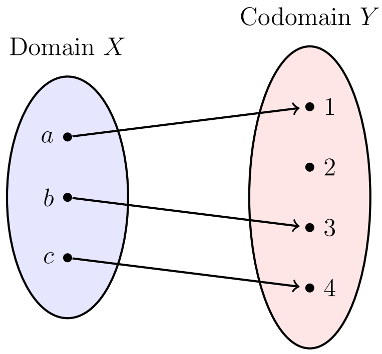
Here is the complete breakdown of how these mappings work mathematically.

Remark A.1 (Syllabus Note)

These concepts are outside the HSC Mathematics Extension 1 and 2 syllabus, but they are very good to know for higher education. In combinatorics, they help formalize counting by pairing elements perfectly between sets. In functions, bijections define exactly when an inverse function exists.

Injective / One-to-one

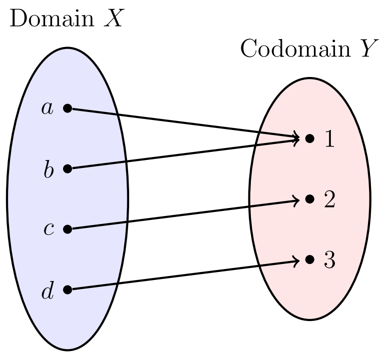
A function is injective if every output is mapped by *at most* one input. In other words, no two different inputs will ever give you the same output.



- **Formal Definition:** If $f(a) = f(b)$, then it must be true that $a = b$.
- **Visualizing it:** Imagine a group of people (domain) and a group of chairs (codomain). An injective mapping means no two people are sitting in the same chair. There might be empty chairs left over, but no chair is double-occupied.
- **Math Example:** $f(x) = 2x$ is injective. Every unique x produces a unique y. However, $f(x) = x^2$ is *not* injective because $f(2) = 4$ and $f(-2) = 4$ (two different inputs hit the same output).

Surjective / Onto

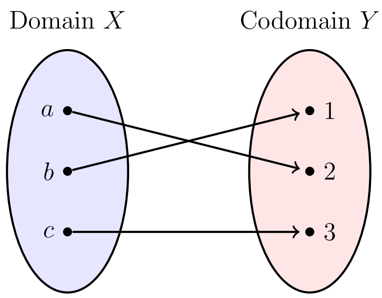
A function is surjective if every possible output in the target set (codomain) is hit by *at least* one input. There are no "leftover" or unreachable outputs.



- **Visualizing it:** Every chair in the room has at least one person sitting in it. (Some chairs might have two people squeezing into them, but no chair is empty).

Bijective / One-to-one correspondence

A function is bijective if it is **both** injective and surjective. Every input maps to a unique output, and every possible output is mapped to by an input.



- **Visualizing it:** It is a perfect pairing. There is exactly one person for every chair, and exactly one chair for every person. No empty chairs, no double-occupied chairs.
- **Why it matters:** Bijective functions are incredibly important in mathematics because they are the **only** functions that have a perfect **inverse function** (f^{-1}). Because the mapping is a perfect two-way street, you can flawlessly travel backward from the output to the exact input.
- **Math Example:** $f(x) = 2x + 1$ (where x is any real number) is a bijection. You can easily find its inverse: $f^{-1}(x) = \frac{x-1}{2}$.

